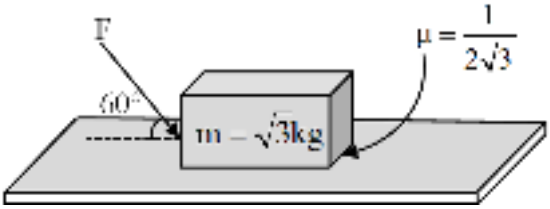
PHYSICS

1) What is the maximum value of the force F such that the block shown in the arrangement, does not move ..... N :-



(1) 20

(2) 10

(3) 12

(4) 15

2) The mass of a ball is 1.76 kg. The mass of 25 such balls is :

(1) 0.44×10^3 kg

(2) 44.0 kg

(3) 44 kg

(4) 44.00 kg

3) Displacement of a particle is given as :

$x = A e^{-kt} \sin (Bt + C)$

which of the following is correct :-

(1) $\left(\frac{B}{k}\right)$ is dimensionless

(2) dimensions of K is $[T^1]$

(3) Dimension of $\frac{A}{k^2}$ is $[LT^{-2}]$

(4) [A] = [C]

4) Arrange the following vernier callipers in increasing order of accuracy :-

(A) 1 M.S.D. = 2 mm 10 V.S.D. = 9 M.S.D.

(B) 1 M.S.D. = 1 mm 10 V.S.D. = 9 M.S.D.

(C) 1 M.S.D. = 2 mm 20 V.S.D. = 19 M.S.D.

(D) 1 M.S.D. = $\frac{1}{2}$ mm 10 V.S.D. = 9 M.S.D.

(1) A < B = C < D

(2) A > B = C > D

(3) B = C < A < D

(4) A < B = D < C

5) When a car is at rest, its driver sees rain drops falling on it vertically. When driving the car with speed v, he sees that rain drops are coming at an angle 60° from the horizontal. On further increasing the speed of the car to (1 + β)v, this angle changes to 45°. The value of β is close to :

(1) 0.41

(2) 0.50

(3) 0.37

(4) 0.73

6) A particle is moving on a straight line having time (t) - velocity (ν) relation as $t = \nu^2 + 2\nu + 1$. Where t is in second and ν is in m/s. Find acceleration of the particle at t = 4 sec.

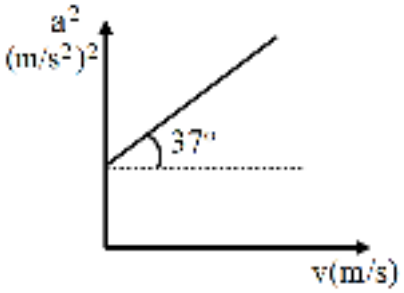
(1) 4 m/s^2

(2) $\frac{1}{4}$ m/s^2

(3) 2 m/s^2

(4) $\frac{1}{2}$ m/s^2

7) In the graph shown between a^2 vs v where a is the acceleration v is the velocity of particle, if rate of change of acceleration is $\frac{3}{x}$ (m/s^3) when velocity of particle will be 5 m/s. Find x.



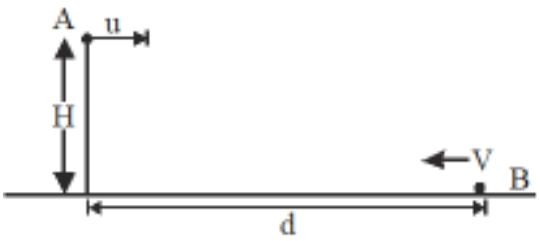
(1) 5

(2) 3

(3) 8

(4) 16

8) Two particles A and B are placed as shown in the figure. The particle A, on the top of tower, is projected horizontally with a velocity u and particle B is projected along the surface towards the tower, simultaneously. If both particles meet each other. Then speed of projection of particle B is



[Ignore any friction]

(1) $d\sqrt{\frac{g}{2H}}-u$

(2) $d\sqrt{\frac{g}{2H}}$

(3) $d\sqrt{\frac{g}{2H}}+u$

(4) u

9) The position of a particle is observed at x = 2m, which was moving in a straight line, having constant acceleration of $2m/s^2$ along positive x-axis. The initial velocity observed is 3m/s in the opposite direction of the acceleration. Match List-I with List-II. All the answers are in SI units.

	List-I		List-II
(P)	The time when the particle comes to rest is	(1)	0.25
(Q)	The minimum speed of the particle is	(2)	3
(R)	The time when the particle comes to the initial position again is	(3)	1.5
(S)	Minimum distance of particle from origin is	(4)	0

(1) P→3, Q→4, R→2, S→1

(2) P→3, Q→4, R→2, S→4

(3) P→1, Q→2, R→3, S→4

(4) P→4, Q→1, R→3, S→2

10) A projectile is thrown with a velocity $10\sqrt{2}$ m/s at 45° with horizontal. The time interval between the moments when speed $\sqrt{125}$ m/s is

(1) 0.5 sec

(2) 1 sec

(3) 1.5 sec

(4) 2 sec

11) A point mass perform straight line motion along positive x-axis. At t = 0 point mass is at A (5,0). It moves such that its velocity is given by $v = \frac{9}{x}$, where x is the x-coordinate of position vector of point mass at a certain time t. Find the time required to move from A to B (13,0) in sec.

(1) 10 sec

(2) 30 sec

(3) 8 sec

(4) None of these

12) A particle is moving on a circular path of radius $\frac{100}{\sqrt{19}}$ m in such a way that magnitude of its velocity varies with time as $v = 2t^2 + t$, where v is velocity in m/s and t is time in s. The acceleration of the particle at t = 2 s is

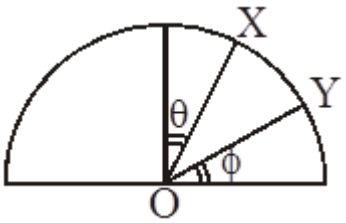
(1) 21 m/s^2

(2) 9 m/s^2

(3) 10 m/s^2

(4) 13.5 m/s^2

13) A particle is released on a vertical smooth semi circular track from point X so that OX makes angle θ from the vertical (see figure). The normal reaction of the track on the particle vanishes at point Y where OY makes angle ϕ with the horizontal. Then :



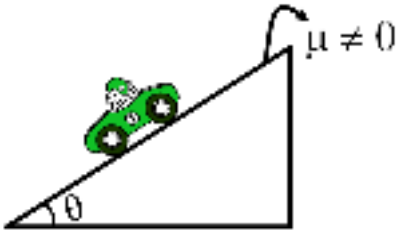
(1) $\sin\phi = \cos\theta$

(2) $\sin\phi = \frac{1}{2}\cos\theta$

(3) $\sin\phi = \frac{3}{4}\cos\theta$

(4) $\sin\phi = \frac{2}{3}\cos\theta$

14) For maximum safe speed of a car, select correct relation for given diagram



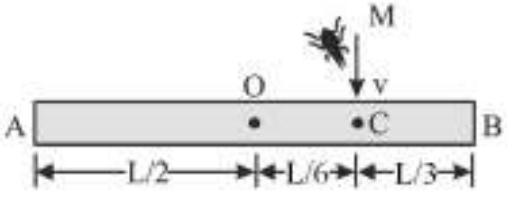
(1) $N\sin\theta + F_L\cos\theta = \frac{mv^2}{r}$

(2) $N\sin\theta - F_L\cos\theta = \frac{mv^2}{r}$

(3) $F_L\cos\theta - N\sin\theta = \frac{mv^2}{r}$

(4) $N\cos\theta + F_L\sin\theta = \frac{mv^2}{r}$

15) A homogeneous rod AB of length L and mass M is pivoted at the center O in such a way that it can rotate freely in the vertical plane. The rod is initially in the horizontal position. An insect of the same mass M falls vertically with speed v on the point C between O & B and stick there. Determine the initial angular velocity in terms of v and L.



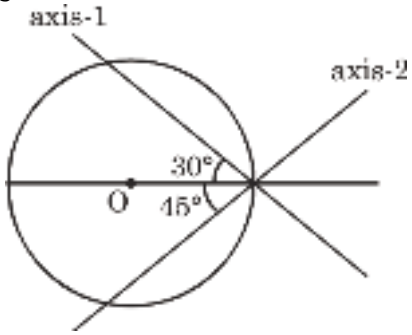
(1) $\frac{12V}{7L}$

(2) $\frac{3V}{2L}$

(3) $\frac{12V}{5L}$

(4) $\frac{9V}{4L}$

16) In figure, the ratio of moment of inertia of a thin uniform disc about axis-2 and axis-1 marked



will be :

(1) 1

(2) 0.5

(3) 0.75

(4) 1.5

17) A small spherical ball is released in a viscous liquid whose density is half that of density of the ball. After attaining terminal speed, ratio of work done by viscous force to work done by gravity on ball during a certain displacement is :-

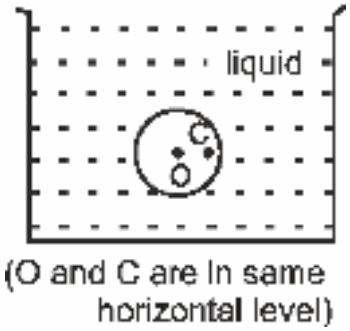
(1) $\frac{1}{2}$

(2) –2

(3) $-\frac{1}{2}$

(4) 2

18) A non-uniform small sphere having average density same as that of the liquid It is released from rest in the position as shown in figure C being the centre of mass and O being the centre of sphere.



Just after it is released :

(1) O moves up

(2) O moves down

(3) C moves left

(4) C moves right

19) A capillary tube is immersed vertically in water and the height of the water column is x. When this arrangement is taken into a mine of depth d, the height of the water column is y. If R is the radius of earth, the ratio $\frac{x}{y}$ is :

(1) $\left(1-\frac{2d}{R}\right)$

(2) $\left(1-\frac{d}{R}\right)$

(3) $\left(\frac{R-d}{R+d}\right)$

(4) $\left(\frac{R+d}{R-d}\right)$

20) When the load on a wire is increased slowly from 2kg to 4kg the elongation increases from 0.6 mm to 1 mm. Than find work done during the extension of the wire.

(1) 14×10^{-3} J

(2) 12×10^{-3} J

(3) 14×10^{3} J

(4) 12×10^{3} J

21) Radius of a capillary is 2×10^{-3} m. A liquid of weight 6.2×10^{-4} N may remain in the capillary. then the surface tension of liquid will be :
(Given the angle of contact is Zero)

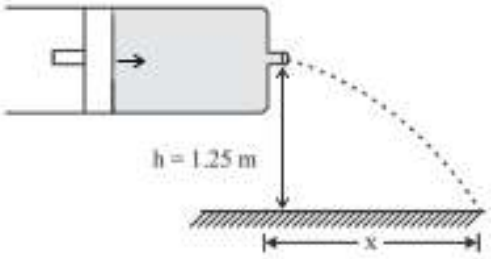
(1) 5×10^{-3} N/m

(2) 5×10^{-2} N/m

(3) 5 N/m

(4) 50 N/m

22) A horizontal tube has two different areas of cross section having diameter of 8mm and 2mm as shown in figure. The range of water falling on horizontal surface, if piston is moving with a constant velocity of 0.25 ms^{-1} is



(h = 1.25 m and g = 10 ms^{-2})

(1) 1 m

(2) 2m

(3) 3m

(4) 4m

23) A ball of mass 1 Kg is moving with velocity $\vec{V}_1 = 3\hat{i} + 4\hat{j}$ collides with the wall and after collision $\vec{V}_2$ is $-2\hat{i} + 6\hat{j}$ which of the following unit vector is perpendicular to wall.

(1) $\frac{1}{5}\left(-3\hat{i} + 4\hat{j}\right)$

(2) $\frac{1}{\sqrt{29}}\left(-5\hat{i} + 2\hat{j}\right)$

(3) $\frac{1}{\sqrt{40}}\left(-2\hat{i} + 6\hat{j}\right)$

(4) $\frac{1}{\sqrt{29}}\left(5\hat{i} - 2\hat{j}\right)$

24) Two uniform solid sphere of equal radii R but mass M and 4M have a centre separation 6R. The two spheres are held fixed. A projectile of mass 'm' is projected from the surface of the sphere of mass M directly towards the centre of the second heavy sphere, then minimum speed of the projectile so that it reaches the surface of the second sphere will be :-

(1) $\sqrt{\frac{5GM}{3R}}$

(2) $\sqrt{\frac{3GM}{5R}}$

(3) $\sqrt{\frac{4GM}{3R}}$

(4) $\sqrt{\frac{2GM}{R}}$

25)

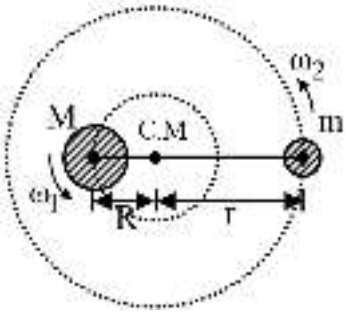


Figure shows two bodies of masses M and m are moving in circles of radii R and r around the centre of mass of system under their mutual gravitational force. If angular velocities are ω_1 and ω_2, time periods are T_1 and T_2 and speeds are v_1 and v_2 for M and m respectively. Then the correct option is :

(1) $T_1 > T_2$

(2) $\omega_1 > \omega_2$

(3) $v_1 = v_2$

(4) $\omega_1 = \omega_2$

26) Four particles of equal masses M moves along a circle of radius R under the action of their

mutual gravitational attraction. The speed of each particle is :-

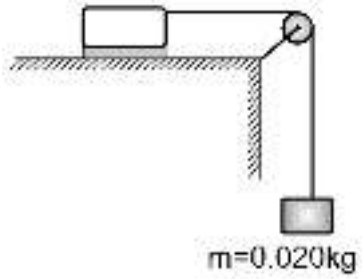
(1) $\left[\frac{GM}{4R}\left(2\sqrt{2}-1\right)\right]^{1/2}$

(2) $\left[\frac{GM}{4R}\left(2\sqrt{2}+1\right)\right]^{1/2}$

(3) $\left[\frac{GM}{R}\left(2\sqrt{2}+1\right)\right]^{1/2}$

(4) $\left[\frac{GM}{R}\left(2\sqrt{2}-1\right)\right]^{1/2}$

27) A metal block of area 0.10 m^2 is connected to a 0.020 kg mass via a string that passes over an ideal pulley as shown in figure. A liquid with a film thickness of 0.30 mm is placed between the block and the table. When system is released the block moves to the right with a small constant speed of



0.090 ms^{-1}, then coefficient of viscosity of liquid is :

(1) $\frac{10}{3} \times 10^{-3}$ Pa s

(2) $\frac{20}{3} \times 10^{-3}$ Pa s

(3) 10^{-2} Pa s

(4) $\frac{40}{3} \times 10^{-3}$ Pa s

28) A sphere of mass m and radius r is projected in a gravity free space with speed v. If coefficient of viscosity of the medium in which it moves is $\frac{1}{6\pi}$, the distance travelled by the body before it stops is :

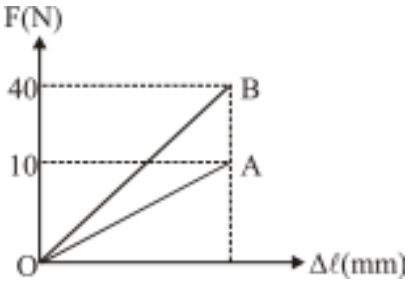
(1) $\frac{mv}{2r}$

(2) $\frac{2mv}{r}$

(3) $\frac{mv}{r}$

(4) $\frac{mv}{4r}$

29) Two wires A and B of same length and made of same material are hanging from the same roof. The figure shows the load F versus extension $\Delta\ell$ graph for the two wires then :-

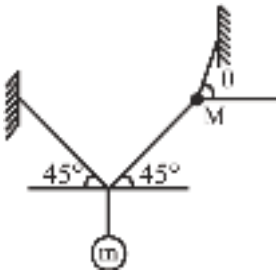


(1) The elasticity of wire A is greater than that of B

(2) The cross-sectional area of wire A is equal to that of B

(3) The ratio of radii of wires A to B is 1 : 2

(4) The ratio of area of cross-section of wires A to B is 4 : 1

30) For equilibrium 

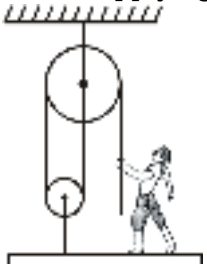
(1) $\tan\theta = 1 + \frac{2M}{m}$

(2) $\tan\theta = 1 + \frac{2m}{M}$

(3) $\tan\theta = 1 + \frac{M}{2m}$

(4) $\tan\theta = 1 + \frac{m}{2M}$

31) A man of mass 60 kg is standing on a platform of mass 40 kg as shown in figure then what force man is applying on platform if he accelerates up with the platform with acceleration of 2 m/s^2:-



(1) 300 N

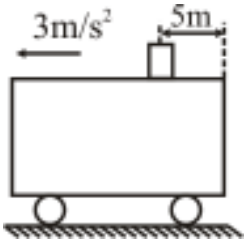
(2) 320 N

(3) 400 N

(4) 480 N

32) 10 kg block is placed 5m from rear end of a trolley coefficient of friction between block & trolley

is 0.2 starting from rest trolley is provided 3 m/s^2 constant acceleration. After what distance moved by trolley block will fall ? (g = 10m/s^2)



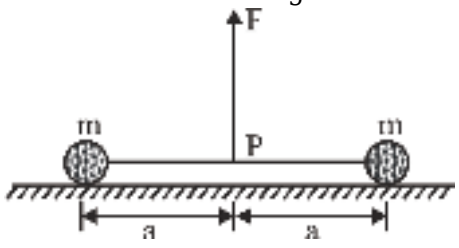
(1) 10 m

(2) 15 m

(3) 20 m

(4) 25 m

33) Two particles of mass 'm' each are tied at the ends of a light string of length 2a. The whole system is kept on a frictionless horizontal surface with the string held tight so that each mass is at a distance 'a' from the centre P (as shown in the figure). Now, the mid-point of the string is pulled vertically upwards with a small but constant force F. As a result, the particles move towards each other on the surface. The magnitude of acceleration, when the separation between them becomes 2x, is :-



(1) $\frac{F}{2m}\frac{a}{\sqrt{a^2-x^2}}$

(2) $\frac{F}{2m}\frac{x}{\sqrt{a^2-x^2}}$

(3) $\frac{F}{2m}\frac{x}{a}$

(4) $\frac{F}{2m}\frac{\sqrt{a^2-x^2}}{x}$

34) The potential energy of a 1 kg particle free to move along the x-axis is given by V(x) = $\left(\frac{x^4}{4}-\frac{x^2}{2}\right)$ J .The total mechanical energy of the particle is 2J. Then the maximum speed (in m/s) is :

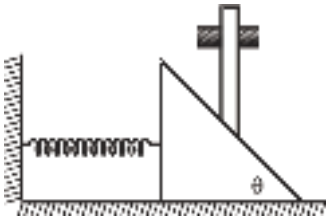
(1) $\frac{3}{\sqrt{2}}$

(2) $\sqrt{2}$

(3) $\frac{1}{\sqrt{2}}$

(4) 2

35) A wedge of mass M fitted with a spring of stiffness 'k' is kept on a smooth horizontal surface. A rod of mass m is kept on the wedge as shown in the figure. System is in equilibrium. Assuming that

all surfaces are smooth, the potential energy stored in the spring is: 

(1) $\frac{mg^2 \tan\theta}{2k}$

(2) $\frac{m^2 g \tan\theta}{2k}$

(3) $\frac{m^2 g^2 \tan\theta}{2k}$

(4) $\frac{m^2 g^2 \tan^2\theta}{k}$

36)

A particle of mass 0.2 kg is moving in one dimension under a force that delivers a constant power 0.5W to the particle. If the initial speed (in ms^{-1}) of the particle is zero, the speed (in ms^{-1}) after 5 s is.

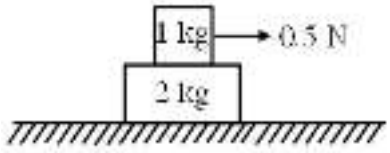
(1) 4 m/s

(2) 5 m/s

(3) 10 m/s

(4) 25 m/s

37) A force of 0.5 N is applied on the upper block as shown in figure. The coefficient of static friction between the two blocks is 0.1 and that between the lower block and the surface is zero. The work done by the lower block on the upper block for a displacement of 3 m of the upper block is :



(1) 1 J

(2) –1 J

(3) 2 J

(4) –2 J

38) If $Q = \frac{X^2}{\sqrt{Y}}$ and ΔX is absolute error in the measurement of X, ΔY is absolute error in the measurement of Y, then absolute error ΔQ in Q is :-

(1) $\frac{4XY\Delta X + X^2\Delta Y}{2Y^{1/2}}$

(2) $\frac{(4XY)\Delta X + X^2\Delta Y}{2Y^{3/2}}$

(3) $2\Delta X + \frac{1}{2}\Delta Y$

(4) $2\Delta X - \frac{1}{2}\Delta Y$

39) The least count of a screw guage is 0.01 mm. The percentage error in measuring 5 mm thickness will be :-

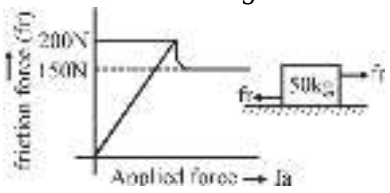
(1) 0.01%

(2) 0.1%

(3) 0.2%

(4) 0.02%

40) The graph between applied force and frictional force for a body of mass 50 kg placed on a horizontal surface is as shown in figure then what will be the value of cofficient of static and kinetic friction will be :-



(1) 0.4, 0.3

(2) 0.3, 0.2

(3) 0.4, 0.2

(4) 0.2, 0.3

41) Two particles of masses m_1 and m_2 in projectile motion have velocities $\vec{v}_1$ and $\vec{v}_2$ respectively at time t = 0. They collide at time t_0. Their velocities become $\vec{v}'_1$ and $\vec{v}'_2$ at time $2t_0$ while still moving in air. The value of $\left|\left(m_1\vec{v}'_1 + m_2\vec{v}'_2\right) - \left(m_1\vec{v}_1 + m_2\vec{v}_2\right)\right|$ is:-

(1) Zero

(2) $(m_1 + m_2)gt_0$

(3) $2(m_1 + m_2)gt_0$

(4) $\frac{1}{2}(m_1 + m_2)gt^0$

42) A satellite is revolving around the earth at a height h from Earth's surface. The satellite explodes into two equal parts, one part has zero velocity after the explosion, and the other part moves in the same direction with double the orbital speed of the original satellite. The maximum height above Earth's surface reached by the part with double velocity is (Neglect the gravitational effects between the fragments):

(1) 3h + 4R

(2) 3R + 4h

(3) 3(R + h)

(4) None of the above

43) A satellite of mass m, initially at rest on the earth, is launched into a circular orbit at a height equal to the radius of the earth. The minimum energy required is :-

(1) $\frac{\sqrt{3}}{4}$ mgR

(2) $\frac{1}{2}$ mgR

(3) $\frac{1}{4}$ mgR

(4) $\frac{3}{4}$ mgR

44) Average torque on a projectile of mass m , initial speed u and angle of projection θ between initial and final positions P and Q , about the point of projection is :

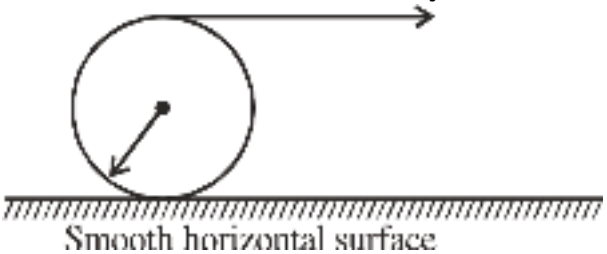
(1) $\frac{mu^2 \sin 2\theta}{2}$

(2) $mu^2 \cos\theta$

(3) $mu^2 \sin\theta$

(4) $\frac{mu^2 \cos\theta}{2}$

45) A thread is tightly wound over a uniform solid cylinder of mass M, which lies on a smooth horizontal surface as shown. The free end of the thread is pulled by horizontal force F as shown. Then acceleration of centre of cylinder at the shown instant is :-



(1) $\frac{F}{2M}$

(2) $\frac{F}{M}$

(3) $\frac{3F}{2M}$

(4) $\frac{2F}{M}$

CHEMISTRY

1) 5g of $CaCO_{3(s)}$ on heating produces $CO_2(g)$ Volume of $CO_2(g)$ at STP is :

(1) 1.12 L

(2) 2.24 L

(3) 4. 48 L

(4) 5.6 L

2) Match the following Column-I to Column-II (Numbers of atoms).

	Column I		**Column II**
(A)	0.5 mole atom of Nitrogen	(P)	3.01×10^{23}
(B)	0.2 mole molecules of Hydrogen	(Q)	2.40×10^{23}
(C)	3.2 g of sulphur	(R)	9.03×10^{23}
(D)	11.2 L of SO_2 gas at NTP	(S)	6.02×10^{22}

(1) (A)–(S); (B)–(Q); (C)–(R); (D)–(P)

(2) (A)–(P); (B)–(S); (C)–(Q); (D)–(R)

(3) (A)–(P); (B)–(Q); (C)–(S); (D)–(R)

(4) (A)–(R); (B)–(P); (C)–(S); (D)–(Q)

3) For the reaction :
$N_2 + 3H_2 \rightleftharpoons 2NH_3$
If volume is increased, then at new equilibrium with respect to previous equilibrium :

(1) Concentration of N_2 will decrease

(2) Concentration of H_2 will decrease

(3) Concentration of NH_3 will decrease

(4) All of these

4) If $\Delta H°$ for formation of CH_4, CO_2 and H_2O are –74.8, – 393.5 kJ/mol and – 285.8 kJ/mol respectively, $\Delta H°$ of combustion of CH_4 is :-

(1) – 890 kJ/mol

(2) – 802 kJ/mol

(3) – 750 kJ/mol

(4) – 650 kJ/mol

5) The heat of neutralization of a strong acid and a strong base is 57.0 kJ mol^{-1}. The heat released when 0.5 mole of HNO_3 is added to 0.20 mole of NaOH is :

(1) 11.4 kJ

(2) 28.5 kJ

(3) 57.0 kJ

(4) 45.6 kJ

6) At constant temperature, the equilibrium constant (K_p) for the decomposition reaction $N_2O_4 \rightleftharpoons 2NO_2$ is expressed by $K_p = (4x^2P)/(1–x^2)$, where P = pressure, x = extent of decomposition. Which statement is true ?

(1) K_p increases with increase of P

(2) K_p increases with decrease of x

(3) K_p increases with decrease of P

(4) K_p remains constant with change in P and x

7) The percentage of Se in peroxidase anhydrase enzyme is 0.5% by weight (atomic weight = 78.4). The minimum molecular weight of the enzyme is :

(1) 1.568×10^4

(2) 3.136×10^4

(3) 15.68

(4) 2.136×10^4

8) Which of the following sets of quantum numbers represents the highest energy of an electron ?

(1) n = 3, □ = 0, m = 0, s = + 1/2

(2) n = 3, □ = 3, m = 1, s = + 1/2

(3) n = 3, □ = 2, m = 1, s = + 1/2

(4) n = 4, □ = 0, m = 0, s = + 1/2

9) Two moles of an ideal gas is mixed with two moles of another identical ideal gas at the same temperature and pressure, ΔS of mixing is :

(1) + ve

(2) Zero

(3) – ve

(4) 2R □n2

10) The equilibrium constant for the reaction 2A + B ⇌ 2C is 40. The equilibrium constant for the reaction C ⇌ A + 1/2 B is :

(1) $\frac{1}{40}$

(2) $\frac{1}{20}$

(3) $\frac{1}{\sqrt{40}}$

(4) $\sqrt{40}$

11) Identify the correct set of wave numbers of first three spectral lines of Balmer series of H-atom. (R = Rydberg const.)

(1) $\frac{7R}{144}, \frac{3R}{16}, \frac{16R}{255}$

(2) $\frac{5R}{36}, \frac{8R}{9}, \frac{15R}{16}$

(3) $\frac{3R}{4}, \frac{3R}{16}, \frac{7R}{144}$

(4) $\frac{5R}{36}, \frac{3R}{16}, \frac{21R}{100}$

12) Which of the following cannot act as an oxidising agent?

(1) N^{3-}

(2) SO_4^{2-}

(3) BrO_3^-

(4) MnO_4^-

13) Solubility of calcium phosphate (molar mass, M) in water is w g per 100 ml at 25°C. Its solubility product at 25°C will be approximately.

(1) $10^7\left(\frac{w}{M}\right)^3$

(2) $10^7\left(\frac{w}{M}\right)^5$

(3) $10^3\left(\frac{w}{M}\right)^5$

(4) $10^5\left(\frac{w}{M}\right)^5$

14) A weak acid HA has degree of dissociation x. Which option gives the correct expression of (pH – pKa) ?

(1) 0

(2) log (1 + 2x)

(3) $\log\left(\frac{1-x}{x}\right)$

(4) $\log\left(\frac{x}{1-x}\right)$

15) Analysis show that nickel oxide consists of nickle ion with 96% ions having d^8 configuration and 4% having d^7 configuration. Which amongst the following best represents the formula of the oxide.

(1) $Ni_{1.02}\ O_{1.00}$

(2) $Ni_{0.96}\ O_{1.00}$

(3) $Ni_{0.98}\ O_{0.98}$

(4) $Ni_{0.98}\ O_{1.00}$

16) An electron in an atom jumps in such a way that its kinetic energy changes from x to $\frac{x}{9}$. The change in its potential energy (magnitude) will be :-

(1) $\frac{x}{9}$

(2) $\frac{16x}{9}$

(3) 9x

(4) $\frac{9x}{16}$

17) The ratio of minimum to maximum wavelengths in Balmer series of H– atom is :

(1) 5 : 9

(2) 5 : 36

(3) 1 : 4

(4) 3 : 4

18) What is the true regarding complete combustion of gaseous isobutane -

(1) $|\Delta H| = |\Delta E|$

(2) $|\Delta H| > |\Delta E|$

(3) $\Delta H = \Delta E = 0$

(4) $|\Delta H| < |\Delta E|$

19) 116 g of a substance upon dissociation reaction yields 7.5 g of hydrogen, 60 g of oxygen and 48.5 g of carbon. The data agrees with which of the following formula

(1) CH_3COOH

(2) CO_2

(3) CH_3COOCH_3

(4) CH_3CHO

20) The de-broglie wavelength of an electron and proton is same. Which has greater momentum ?

(1) Electron

(2) Same

(3) Proton

(4) Cannot be determined

21) For an ideal gas undergoing isothermal expansion, which of the following is correct.

(1) $\Delta U > 0$

(2) $\Delta U < 0$

(3) $\Delta U = 0$

(4) $\Delta H \neq 0$

22) In the reaction
$KMnO_4 + HCl \rightarrow KCl + MnCl_2 + Cl_2 + H_2O$
The oxidising agent is –

(1) HCl

(2) H_2O

(3) $KMnO_4$

(4) $MnCl_2$

23) The maximum number of spectral lines emitted in IR region when electrons in hydrogen atom falls from n = 6 to ground state is :

(1) 4

(2) 6

(3) 10

(4) 15

24) Standard enthalpy of formation of $CO_{2(g)}$ is – 393 KJ/mol. This corresponds to ?

(1) $CO_{2(g)}$ formation from $CO_{(g)}$

(2) Combustion of graphite(s)

(3) $CO_{2(g)}$ formation from elements in their standard states

(4) Both (2) & (3)

25) For complete combustion of methanol

$$CH_3OH(\ell) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(\ell)$$

the amount of heat produced as measured by bomb calorimeter is 726 kJ/mol at 27°C. The enthalpy of combustion of methanol is-

(1) – 521 KJ/mol

(2) – 595 KJ/mol

(3) – 450 KJ/mol

(4) – 727 KJ/mol

26) Let us consider a reversible reaction at temperature T. In this reaction, both ΔH and ΔS were observed to have positive values. If the equilibrium temperature is T_e, then the reaction becomes spontaneous at -

(1) $T = T_e$

(2) $T_e > T$

(3) $T > T_e$

(4) $T_e = 5T$

27) An aqueous solution of HCl with pH 1.0 is diluted by adding equal volume of water (ignoring dissociation of water) then pH of HCl solution would be
(Given log2 = 0.30)

(1) reduce to 0.5

(2) increase to 1.3

(3) remain same

(4) increase to 2

28) Cation 'X' has pseudo inert gas configuration and cation 'Y' has (18 + $2e^-$) gas configuration (both are of same atom) then **INCORRECT** statement is :

(1) Cation 'Y' has two more electron than the cation 'X'

(2) Positive charge on cation 'Y' must be more than cation 'X'

(3) Cation 'X' needed 8 more electron to convert into noble gas configuration

(4) Cation 'Y' needed 6 more electrons to convert into noble gas configuration

29) If internuclear axis is x-axis then HOMO orbital(s) of O_2 molecule is..

(1) σ^*2p_x

(2) π^*2p_z, π^*2p_y

(3) π^*2p_y, π^*2p_x

(4) $\pi 2p_z$, $\pi 2p_y$

30) Which of the following matching is correct if s-p intermixing is not considered.

(1) O_2 is dimagnetic, bond order = 2

(2) C_2 is dimagnetic, bond order = 2

(3) B_2 is paramagnetic, bond order = 1

(4) F_2 is dimagnetic, bond order = 1

31) In ICl_2^+ , ICl_2^- and ICl_4^- sum of bond pair and lone pair on each iodine atom are respectively -

(1) 2, 2 and 4

(2) 2, 3 and 2

(3) 4, 5 and 4

(4) 4, 5 and 6

32) Bonding in which of the following diatomic molecule(s) become(s) stronger, on the basis of MO Theory, by removal of an electron ?

(A) NO (B) N_2 (C) O_2

(D) C_2 (E) B_2

Choose the most appropriate answer from the options given below :-

(1) (A), (B), (C) only

(2) (B), (C), (E) only

(3) (A), (C) only

(4) (D) only

33) Identify the correct match:

(i)	XeF_2	(a)	Central atom has sp^3 hybridization and angular geometry

(ii)	N_3^-	(b)	Central atom has sp^3d^2 hybridization and octahedral
(iii)	PCl_5(s) anion	(c)	Central atom has sp hybridization and linear geometry
(iv)	I_2Cl_6(l) cation	(d)	Central atom has sp^3d hybridization and linear geometry

(1) (i-a), (ii-b), (iii-c), (iv-d)

(2) (i-d), (ii-b), (iii-a), (iv-c)

(3) (i-b), (ii-c), (iii-a), (iv-d)

(4) (i-d), (ii-c), (iii-b), (iv-a)

34) The dissolution of ionic compounds involves :

(1) Evolution of heat

(2) Weakening of attractive forces

(3) Dissociation into ions

(4) All of these

35) Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : Dipole-dipole interactions are the only non-covalent interactions, resulting in hydrogen bond formation.
Reason R : Fluorine is the most electronegative element and hydrogen bonds in HF are symmetrical.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) A is false but R is true

(2) Both A and R are true and R is the correct explanation of A

(3) A is true R is false

(4) Both A and R are true but R is NOT the correct explanation of A

36) According to VBT molecule got stability by :-

(1) Completing it's octet

(2) Hybridisation

(3) Pairing it's unpaired e^-

(4) Donating e^-

37) Choose the molecules in which hybridisation occurs in the ground state ?
(i) BCl_3 (ii) NH_3
(iii) PCl_3 (iv) BeF_2
The correct answer is -

(1) i, ii, iv

(2) i, ii, iii

(3) ii, iii

(4) iii, iv

38) Electron gain enthalpy will be positive in

(1) O^{-2} is formed from O^{-1}

(2) O^{-1} is formed from O

(3) S^{-1} is formed from S

(4) Na^{-} is formed from Na

39) Which one of the following statements is incorrect in relation to ionization enthalpy?

(1) Ionization enthalpy increases for the removing each successive electron.

(2) The greatest increase in ionization enthalpy is experienced on removal of electron from core noble gas configuration.

(3) End of valence electrons is marked by a big jump in ionization enthalpy.

(4) Removal of electron from orbitals bearing lower n value is easier than from orbital having higher n value.

40)

Element	$I.E._{I}$	$I.E._{II}$	$I.E._{eg}$
P	520	7300	–60
Q	419	1051	–48
R	1681	3374	–328
S	1008	1846	–295
T	2372	5251	+48

Which of the following statement is not correct for above elements ?

(1) P is the metal which form halide of the formula MX

(2) S is the less reactive non metal

(3) T is the least reactive element

(4) Q is a non-metal and does not form halide of the formula MX_2

41) Which statement is wrong statement for the long form of periodic table :-

(1) Number of periods are 7 and groups are 18

(2) No. of valence shell electrons in a period are same

(3) IIIrd B group contains 32 elements

(4) Lanthanides and actinides are placed in same group

42) The best reason to account for the general tendency of atomic size to decrease as the atomic numbers increase within a period of the periodic table is the fact that

(1) Outer electrons repel inner electrons

(2) Closer packing among the nuclear particles is achieved

(3) The number of neutrons increases

(4) The increasing nuclear charge exerts a greater attractive force on the electrons

43) Which of the following order of radius is not correct ?

(1) Cr < Mo $\simeq$ W

(2) Sc < Y $\simeq$ La

(3) Fe $\simeq$ Co $\simeq$ Ni

(4) Al $\simeq$ Ga

44) **Statement-1 :** order of ionisation energy F > Cl > Cl^- > F^-
Statement-2 : Size Cl^- is smaller than F^-

(1) Statement (1) & (2) both are correct.

(2) Statement (1) & (2) both are incorrect.

(3) Statement (1) is correct but statement (2) is incorrect

(4) Statement (1) is incorrect but statement (2) is correct

45) Identify the **incorrect** order.

(1) $N_2O_5 < P_2O_5 < As_2O_5 < Sb_2O_5$ (Acidic)

(2) $Na_2O > MgO > B_2O_3 > CO_2$ (Basic)

(3) $Li_2O < Na_2O < K_2O < Rb_2O < Cs_2O$ (Basic)

(4) $HClO_4 > HClO_3 > HClO_2 > HOCl$ (Acidic)

BIOLOGY

1) Which of the following option represents correct matching pairs ?

	Category of enzyme		**Example**
(A)	Oxidoreductase	(i)	Aldolase
(B)	Lyases	(ii)	Citrate synthase
(C)	Isomerases	(iii)	Dehydrogenase
(D)	Ligases	(iv)	Mutase

(1) A = (iii), B = (iv), C = (ii), D = (i)

(2) A = (iv), B = (iii), C = (i), D = (ii)

(3) A = (iii), B = (i), C = (ii), D = (iv)

(4) A = (iii), B = (i), C = (iv), D = (ii)

2) How many of following statement/statements is/are incorrect ?

(i) An active site of the enzyme is crevice or pocket into which the substrate fits.
(ii) Both enzyme and inorganic catalyst work efficiency at high temperature
(iii) Thermal stability is an important characteristic of almost all enzymes
(iv) Almost all enzymes are proteins, there are some nucleic acid which behaves like enzyme.

(1) One
(2) Two
(3) Three
(4) Four

3) Which of the following statement is incorrect ?

(1) Only right handed helices are observed in proteins.
(2) Haemoglobin consists of 4 sub units 2α & 2β.
(3) Tertiary configuration is an assembly of more than one polypeptides.
(4) Secondary configuration of protein is insoluble in water.

4) A tetrapeptide made up of 4 amino acids. If molecular weight of one amino acid is 120 Da then what will be the molecular weight of that tetrapeptide ?

(1) 480 Da
(2) 426 Da
(3) 462 Da
(4) 444 Da

5) Peptide bond is formed when the :-

(1) Carboxyl group of one amino acid reacts with the carboxyl group of the next amino acid.
(2) Amino group of one amino acid reacts with the amino group of the next amino acid.
(3) Carboxyl group of one amino acid reacts with amino group of the next amino acid.
(4) Amino group of one amino acid reacts with aldehyde group of the next amino acid.

6) **Statement-I :** Outer membrane of chloroplast is more permeable than inner membrane of mitochondria.
Statement-II : Mitochondria unless specifically stained are not easily visible under the microscope.

(1) Both statements are incorrect.
(2) Both statements are correct.
(3) Only statement-I is correct.
(4) Only statement-II is correct.

7) Match the column-I with column-II and select the correct option-

	Column-I		**Column-II**
(A)	Chromatin	(i)	Present only in plant cells
(B)	Centromere	(ii)	Loose & indistinct network of nucleoprotein fibres.

(C)	Metacentric	(iii)	Centromere is located at mid of the chromosome
(D)	Kinetochore	(iv)	Primary constriction
(E)	Microbodies	(v)	Disc shaped structure
		(vi)	Minute vesicle bounded by membrane

Options -

(1) A - iv, B - ii, C - iii, D - v, E - vi

(2) A - ii, B - iv, C - iii, D - v, E - vi

(3) A - iv, B - i, C - iii, D - v, E - vi

(4) A - v, B - i, C - iii, D - iv, E - ii

8) **Statement-I :** Prokaryotic cells are represented by bacteria, blue-green algae, red algae and mycoplasma.
Statement-II : Prokaryotic cells are generally larger and multiply more rapidly than the eukaryotic cells.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II are incorrect.

9) Read the following statements and choose the correct option.
Statement - I : The centrioles form the axoneme of cilia and flagella and spindle fibres that give rise to spindle apparatus during cell division in plant cells.
Statement - II : The cytoskeleton in a cell are involved in many functions such as mechanical support, motility and maintenance of the shape of the cell.

(1) Both Statements are correct.

(2) Both Statements are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

10) **Assertion :** In mitochondria and plastid small, double stranded, circular DNA and ribosomes are present.
Reason : Ribosome of mitochondria and plastid serve as site for protein synthesis.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

11) Cilia and flagella are hair like outgrowths of ____ :

(1) Cell wall

(2) Cell membrane

(3) Cytoskeleton

(4) Golgi complex

12) Which of the following is/are found in both prokaryotic and eukaryotic cells. ?
(A) Semifluid matrix cytoplasm
(B) Selectively permeable plasma memberane.
(C) 70 s Ribosome
(D) Membrane bound nucleus
(E) Genetic material DNA
(F) Chromatophores

(1) A, B, C, & D
(2) A, B, D, & E
(3) B, D, E, & F
(4) A, B, C, & E

13) **Assertion :-** A telocentric chromosome has two unequal arms.
Reason :- The centromere is situated close to its end, in case of acrocentric chromosome.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Assertion is false but Reason is true.
(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

14) Match List-I with List-II

List-I		**List-II**	
(a)	Stroma of chloroplast	(i)	60s + 40 s subunit
(b)	Cytoskeleton	(ii)	Flat disc-shaped cisternae
(c)	80s Ribosome	(iii)	Enzymes required for synthesis of proteins
(d)	Golgi apparatus	(iv)	Proteinaceous filament

(1) a-iii, b-i, c-iv, d-ii
(2) a-ii, b-iii, c-iv, d-i
(3) a-iv, b-iiii, c-i, d-ii
(4) a-iii, b-iv, c-i, d-ii

15) In which type of chromosome p arm is very short compared to the q arm ?

(1) Metacentric
(2) Telocentric
(3) Sub meta centric
(4) Acrocentric

16) If the 2n = 28 in plant cell then what is possible in metaphase–I of meiosis :

(1) 28 Bivalent, 56 Centromere, 112 Chromatids

(2) 14 Bivalent, 28 Centromere, 28 Chromatids

(3) 14 Bivalent, 28 Centromere, 56 Chromatids

(4) 28 Bivalent, 56 Centromere, 56 Chromatids

17) Match the following correctly -

(a)	Formation of synaptonemal complex	(i)	Diplotene
(b)	Appearance of recombination nodule	(ii)	Diakinesis
(c)	Dissolution of synaptonemal complex	(iii)	Zygotene
(d)	Terminalisation of chiasmata	(iv)	Pachytene

(1) (a-iii), (b-ii), (c-i), (d-iv)

(2) (a-iii), (b-iv), (c-i), (d-ii)

(3) (a-iii), (b-i), (c-iv), (d-ii)

(4) (a-i), (b-ii), (c-iii), (d-iv)

18) Match the column-I with column-II and select the correct option.

	Column-I		**Column-II**
(a)	Chromosome decondense and loose their individuality	(I)	Metaphase
(b)	Condensation of chromosomes are completed	(II)	Telophase
(c)	Chromatids move to opposite poles	(III)	Prophase
(d)	Mitotic apparatus first time appears	(IV)	Anaphase

(1) (a)-I, (b)-II, (c)-III, (d)-IV

(2) (a)-I, (b)-III, (c)-II, (d)-IV

(3) (a)-II, (b)-I, (c)-IV, (d)-III

(4) (a)-II, (b)-III, (c)-IV, (d)-I

19) Find the incorrect match :

(1) Karyokinesis - nuclear division

(2) Mitosis - equational division

(3) Meiosis - gametogenesis

(4) Interkinesis - DNA replication

20) After meiosis four daughter cells are formed from one mother cell. If each daughter cell has 8 chromosomes then what will be the number of bivalent in Prophase–I of the same mother cell?

(1) 8
(2) 16
(3) 24
(4) 32

21) If a cell has 15 bivalents in zygotene stage, how many kinetochores are found in it ?

(1) 30
(2) 60
(3) 15
(4) 45

22) One molecule of enzyme converts the 5 molecules of substrate into product in 5 minutes. 30 molecules of substrate and two molecules of enzyme are mixed in a test tube. After 10 minutes the test tube will have :-

(1) Products and 10 catalysed molecules of substrate
(2) Only product molecules
(3) 10 uncatalysed molecules of substrate, enzymes and product molecules
(4) Products and enzymes molecules

23) Read the following statements : (A) Photoperiod affects reproduction in seasonal breeders, both plants and animals.
(B) Increase in body mass is considered as reproduction.
(C) Metabolism is a defining feature of all living organism without exception.
(D) Self consciousness is a defining property of every living organism. Choose the correct statements

(1) A and B
(2) B and C
(3) A and C
(4) B and D

24) What is illustrated by statement "nothing lives forever, yet life continues"?

(1) Growth
(2) Locomotion
(3) Metabolism
(4) Reproduction

25) Identify the correct match from the column-I, II and III.

Column-I	**Column-II**	**Column-III**

(I)	Wheat	(a)	*Homo*	(i)	Monoco-tyledonae
(II)	Man	(b)	*Musca*	(ii)	Dico-tyledonae
(III)	Mango	(c)	*Mangifera*	(iii)	Insecta
(IV)	Housefly	(d)	*Triticum*	(iv)	Mammalia

(1) I-d-i, II-a-iv, III-c-ii, IV-b-iii

(2) I-d-i, II-c-ii, III-a-iv, IV-b-iii

(3) I-d-ii, II-c-i, III-a-iv, IV-b-iii

(4) I-a-iv, II-d-i, III-b-iii, IV-c-ii

26) Study the following statement and give the answer:-
(A) Bacteria are sole members of the kingdom monera
(B) Bacteria are most abundant micro-organisms
(C) A handful of soil contain thousands of bacteria
(D) In bacteria, mesosome help in respiration
(E) Bacteria show most extensive metabolic diversity

(1) Only A, B & E are correct

(2) Only A, C & E are correct

(3) Only A, B, D & E are correct

(4) A, B, C, D and E are correct

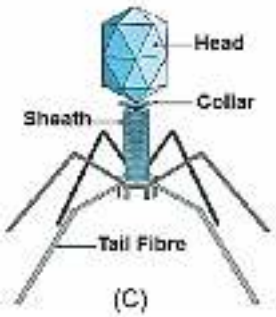
27) Identify figures A, B, C and D :



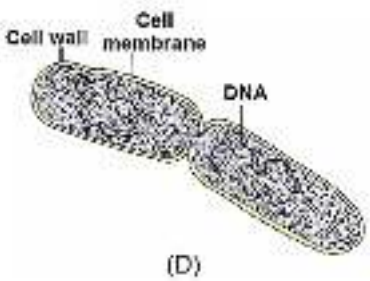
(A)



(B)



(C)



(D)

(1) *A = Agaricus, B = Paramoecium,*
C = Bacteriophage, *D - Anabaena*

(2) *A= Aspergillus, B = Euglena,*
C = TMV, D = Dividing bacteria

(3) *A = Aspergillus, B = Paramoecium,*
C = Bacteriophage, *D* = Dividing bacteria

(4) *A = Mucor, B = Euglena,*
C = TMV, *D = Nostoc*

28) How many of the following fungi have septate and Branched Mycelium with absence of asexual spores
Neurospora, Alternaria, Claviceps, agaricus, Colletotrichum, Penicillium, Trichoderma, Aspergillus, Mucor, Rhizopus and Albugo

(1) 8
(2) 3
(3) 1
(4) All

29) Which one of the following is not the characteristic feature of cyanobacteria ?

(1) They can form colonies
(2) They can fix atmospheric N_2
(3) They are eukaryotes
(4) They form blooms in polluted water bodies.

30) The protistans having protein rich layer which makes their body flexible are :-

(1) Slime moulds
(2) Chrysophytes
(3) Dinoflagellates
(4) Euglenoids

31) Mycelium as continous tubes filled with multinucleated (Coenocytic) cytoplasm is found in :

(1) *Saccharomyces* and *Penicillium*
(2) *Rhizopus, Mucor* and *Albugo*
(3) *Rhizopus, Ustilago* and *Puccinia*
(4) *Alternaria, Colletotrichum* and *Albugo*

32) Prion is :

(1) A nucleocapsid
(2) A low molecular weight free RNA
(3) A protozoan
(4) An abnormally folded protein

33) Choose the wrong statement :

(1) *Neurospora* is used in the study of biochemical genetics
(2) Morels and truffles are poisonous mushrooms
(3) Yeast is unicellular and useful in fermentation
(4) *Penicillium* is multicellular and produces antibiotics

34) The infectious agent that causes potato spindal tuber disease was discovered by ?

(1) D.J. Ivanowsky

(2) W.M. Stanley

(3) T.O. Diener

(4) M. W Beijerinck

35) **Assertion :** Fungi show great diversity in morphology and habitat.
Reason : Sexual cycle in fungi involves three steps-
(i) Plasmogamy
(ii) Karyogamy
(iii) Meiosis

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

36) **Assertion:** Methanogens are present in the gut of several ruminant animals.
Reason: Methanogens are responsible for the production of methane from the dung of ruminants animals.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

37) **Assertion:** Archaebacteria survive in extreme conditions.
Reason: Archaebacteria differ from other bacteria in having a different (complex) cell wall structure.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

38) **Assertion :** In gymnosperms, endosperm is haploid in nature.
Reason : In gymnosperms formation of endosperm is pre-fertilisation event and it is formed by functional megaspose.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

39) Read the following statements and identify the characters related to the fern shown in the



diagram : (a) Homosporous plant
(b) Heterosporous plant
(c) Precursor to the seed habit
(d) Belong to sphenopsida class
(e) Non-motile male gamete

(1) (b) and (c) only
(2) (a), (b) and (c) only
(3) (a), (c) and (d) only
(4) (c), (d) and (e) only

40) Which of the following is a dioecious Bryophyta ?

(1) *Marchantia*
(2) *Funaria*
(3) *Sphagnum*
(4) *Pteris*

41) Ovule of gymnosperm is equivalent to :

(1) Archegonium
(2) Endopserm
(3) Microsporangium
(4) Megasporangium

42) **Statement-I :-** The plant body of bryophytes is more differentiated than that of algae.
Statement-II :- The sex organs in bryophytes are unicellular.

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

43) Read the following statement and select the correct, option.
(A) In conifers needle like leaves reduce the surface area.
(B) Most of gymnosperm are heterosporous.
(C) Reduced male gametophyte in gymnosperm is called a pollen grain.
(D) Megaspore mother cell divides mitotically to form four megaspore.

(1) 2 - correct, 2 - incorrect

(2) 3 - correct, 1 - incorrect

(3) 1 - correct, 3 - incorrect

(4) 0 - correct, 4 - incorrect

44) **Statement (A) :** The pteridophytes include horsetails and ferns.
Statement (B): The pteridophytes are classified into four classes : Psilopsida, lycopsida, sphenopsida, pteropsida.

(1) **Statement (A)** and **Statement (B)** both are incorrect.

(2) **Statement (A)** and **Statement (B)** both are correct.

(3) **Statement (A)** is correct but **Statement (B)** is incorrect.

(4) **Statement (A)** is incorrect but **Statement (B)** is correct.

45) Which of the following is not correctly matched ?

(1) *Chlamydomonas* - Unicellular, flagellated

(2) *Laminaria* - used as food

(3) *Ulothrix* - Filamentous form

(4) *Volvox* - Colonial form, Non - Flagellated.

46) In which animal teeth are modified placoid scale.

(1) Neophron

(2) Naja

(3) Labeo

(4) Torpedo

47) Amphioxus or lancelet is a :-

(1) Cephalochordate with notochord only in its tail

(2) Urochordate with notochord only in its tail

(3) Cephalochordate with notochord extended from head to tail

(4) Urochordate with notochord extended from head to tail

48) Match column I & II and select appropriate option.

	Column-I		**Column-II**
(i)	Sea anemone	(a)	Pennatula
(ii)	Sea fan	(b)	Meandrina
(iii)	Brain coral	(c)	Gorgonia
(iv)	Sea pen	(d)	Adamsia

(1) (i)-d, (ii)-c, (iii)-b, (iv)-a

(2) (i)-a, (ii)-b, (iii)-c, (iv)-d

(3) (i)-d, (ii)-b, (iii)-c, (iv)-a

(4) (i)-b, (ii)-c, (iii)-a, (iv)-d

49) **Statement 1** : In open type circulatory system blood is pumped out of the heart and cells and tissues are directly bathed in it.
Statement 2 : In closed type circulatory system the blood is circulated through series of vessels of same diameters.

(1) Both Statement 1 and 2 are incorrect.

(2) Statement 1 is incorrect and Statement 2 is correct.

(3) Statement 1 is correct and Statement 2 is incorrect.

(4) Both Statement 1 and 2 are correct.

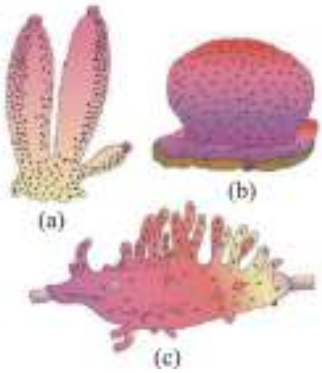
50) Select the correct statement with regard to echinoderms.

(1) They are triploblastic and acoelomate animals.

(2) Their digestive system is complete with mouth on upper side & anus on lower side.

(3) Excretory system is absent in Echinoderms.

(4) Fertilization in echinoderms is external and development is direct.



51) Identify the following animal a, b and c :-

	a	b	c
(1)	*Sycon*	*Spongilla*	*Euspongia*
(2)	*Euspongia*	*Sycon*	*Spongilla*
(3)	*Spongilla*	*Sycon*	*Euspongia*
(4)	*Sycon*	*Euspongia*	*Spongilla*

(1) 1

(2) 2

(3) 3

(4) 4

52) Select correct statement only about reptiles.

(1) They are mostly viviparous and development is direct.

(2) Reptiles are homeothermic or warm blooded.

(3) They are mostly aquatic animals and their body is covered by dry and cornified skin, epidermal scales or sculles.

(4) Snakes and lizards shed their scales as skin cast.

53) Match column-I with column-II and select correct answer from the options given below :-

	Column-I (Animal)		**Column-II (Common name)**
a	Hyla	i	Toad
b	Ichthyophis	ii	Salamandara
c	Bufo	iii	Limbless amphibia
d	Salamander	iv	Tree frog

Options :-

(1) a-i, b-ii, c-iii, d-iv

(2) a-ii,b-iii,c-iv, d-i

(3) a-iv,b-i, c-ii,d-iii

(4) a-iv,b-iii, c-i,d-ii

54) Given below are two statements one is labelled as assertion (A) and another is labelled as reason (R).

Assertion (A) :- Amphibians are warm blooded, so they undergo hibernation or Aestivation to prevent themselves from extreme cold and heat.

Reason (R) :- The amphibian skin is highly thin, moist, vascular and glandular with scales.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is incorrect.

(4) Both (A) and (R) are incorrect.

55) In which one of the following, the genus name, its two characters and its phylum are not correctly matched, whereas the remaining three are correct ?

	Genus Name	**Two characters**	**Phylum**
(1)	*Sycon*	(a) Pore bearing (b) Water Canal system	Porifera
(2)	*Periplaneta*	(a) Jointed appendages (b) Chitinous exoskeleton	Arthropoda
(3)	*Pila*	(a) Body segmented (b) Mouth with Radula	Mollusca
(4)	*Asterias*	(a) Spiny skinned (b) Water vascular system	Echinodermata

(1) 1

(2) 2

(3) 3

(4) 4

56) Match the column-I and column-II :

	Column-I		Column-II
1.	White fibrous connective tissue	a.	Endosteum
2.	Yellow fibrous connective tissue	b.	Tendon
3.	Adipose connective tissue	c.	Tela subcutenea
4.	Areolar connective tissue	d.	Ligament
5.	Reticular fibrous connective tissue	e.	Hump of camel

(1) 1 – b, 2 – d, 3 – e, 4 –c, 5 –a

(2) 1 – c, 2 – a, 3 – d, 4 –b, 5 –e

(3) 1 – b, 2 – a, 3 – e, 4 –c, 5 –d

(4) 1 – b, 2 – c, 3 – a, 4 –d, 5 –e

57) Which cartilage is present in pubic symphysis?

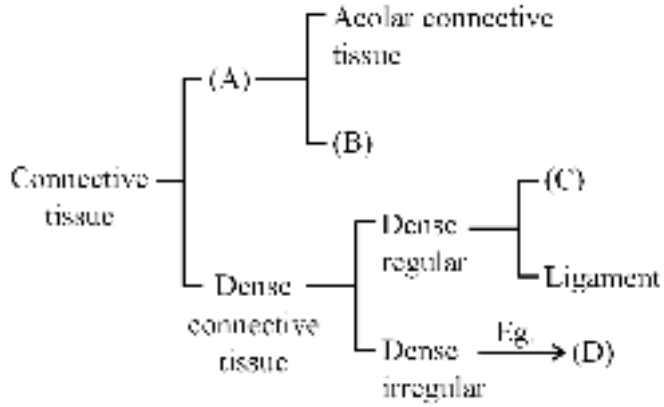
(1) Hyaline cartilage

(2) Elastic cartilage

(3) Calcified cartilage

(4) White fibrous cartilage

58) Read the following flow chart carefully and select the correct option for A, B, C and D



respectively :-

(1) Adipose connective tissue, loose connective tissue, in skin, tendon

(2) Loose connective tissue, adipose connective tissue, tendon, in skin

(3) Loose connective tissue, adipose connective tissue, in skin, tendon

(4) Specialised connective tissue, adipose connective tissue, in skin, tendon

59) Refer the given figure and identify the correct characteristic feature :-



(a) It is a type of loose connective tissue
(b) The cells are specialised to store fats
(c) It is found beneath the skin.

(1) 'a' and 'b'

(2) 'a' and 'c'

(3) 'b' and 'c'

(4) All a, b and c

60) **Assertion :** 6-8 small and tubular, blind structures called hepatic caecae is present at the junction of foregut and midgut.
Reason : Chemical digestion in cockroach starts in midgut.

(1) Both assertion and reason are true and reason is correct explanation of assertion.

(2) Both assertion and reason are true and reason is not correct explanation of assertion.

(3) Assertion is true but reason is false

(4) Assertion is false but reason is true.

61) Choose correct statements about cockroach.
(a) Hind gut is broader than midgut
(b) Hind gut and fore gut is lined with cuticle
(c) Rectum opens out through cloaca
(d) Blood is colourless and also called as haemolymph

(1) (a), (c) and (d)

(2) (a), (b) and (c)

(3) (a), (b) and (d)

(4) Only (c)

62) Genital pouch of male cockroach is dorsally bounded by __(A)__ and __(B)__ and ventrally by the __(C)__
Select the option in which (A), (B) and (C) are given correctly.

	(A)	**(B)**	**(C)**
(1)	9th tergum	10th tergum	9th sternum
(2)	7th sternum	8th sternum	9th sternum
(3)	7th tergum	8th tergum	9th sternum
(4)	9th tergum	10th tergum	10th sternum

(1) 1

(2) 2

(3) 3

(4) 4

63) Which of the following is correctly matched regarding frog :

(1) Copulatory pad - Helps in insemination.

(2) Mesorchium - Single fold of peritoneum.

(3) Hind brain - Cerebrum and medulla oblongata.

(4) RBC - Biconvex and nucleated.

64) Which type of respiration occurs in frog during hibernation and aestivation ?

(1) Both cutaneous and pulmonary

(2) Both cutaneous and bucco-pharyngeal

(3) Only cutaneous

(4) Only pulmonary

65) Read the following four statements (A – D) regarding to amphibians :-
(A) Amphibians can live in aquatic as well as terrestrial habitats
(B) Body is divided into head, neck, trunk and tail
(C) Tympanum represent the ear
(D) The eyes have eye lids
How many of the above statements are correct?

(1) Two

(2) Four

(3) One

(4) Three

66) How many statement are not correct about frog.
(1) Frog is poikilothermal
(2) Frog do not show sexual dimorphism
(3) External ear is absent in frog
(4) Forelimbs & hind limbs of frog have 5 digit & 4 digit respectively.

(1) Two

(2) One

(3) Three

(4) Four

67) In a standard ECG which of the following positions of lead is not correct ?

(1) Right wrist

(2) Left wrist

(3) Right ankle

(4) Left ankle

68) Why there is delay in processing the electrical signal from the AV-node to the AV-bundle ?

(1) To ensure the both atria contract at same time.
(2) To ensure the both ventricles contract at same time.
(3) To ensure the opening of atrio-ventricular valves.
(4) To allow the atria to complete their contraction prior to ventricular contraction.

69) Match the following columns :

(A) Blood group		**(B) Antigen**	
(I)	A	(a)	A & B both
(II)	B	(b)	A
(III)	AB	(c)	nil
(IV)	O	(d)	B

(1) I-b, II-d, III-a, IV-c
(2) I-b, II-d, III-c, IV-a
(3) I-d, II-b, III-a, IV-c
(4) I-b, II-a, III-c, IV-d

70) Match **List-I** with **List-II**

List-I (ECG)		**List-II (Electrical activity of heart)**	
(A)	P-wave	(I)	Depolarisation of ventricles
(B)	QRS complex	(II)	End of systole
(C)	T wave	(III)	Depolarisation of atria
(D)	End of T wave	(IV)	Repolarisation of ventricles

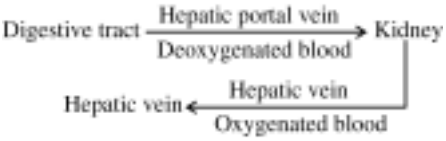
Choose the **correct** answer from the options given below :
(1) A-(IV), B-(I), C-(III), D-(II)
(2) A-(I), B-(IV), C-(III), D-(II)
(3) A-(IV), B-(III), C-(I), D-(II)
(4) A-(III), B-(I), C-(IV), D-(II)

71) Inadequate supply of oxygen to heart muscles leads to a symptom of acute chest pain. This disorder of the circulatory system is identified as :

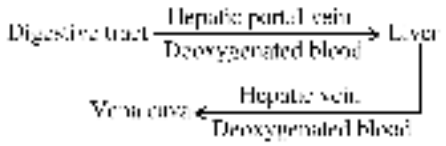
(1) Cardiac arrest
(2) Heart failure
(3) Coronary Heart Disease
(4) Angina pectoris

72) Choose correct option for hepatic portal circulation.

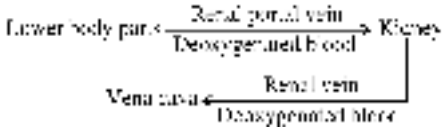
(1)



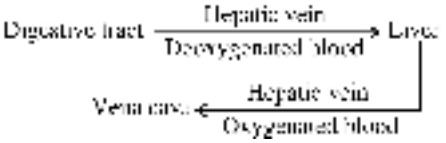
(2)



(3)



(4)



73) Match the respiratory disorder with its description -

Column-A		Column-B	
(I)	Asthma	(a)	Chronic disorder where alveolar walls are damaged
(II)	Emphysema	(b)	Difficulty in breathing causing wheezing
(III)	Occupational respiratory disorder	(c)	Infection and inflammation of respiratory tract
(IV)	Bronchitis	(d)	Inflammation leading to fibrosis due to dust exposure

(1) (I)-(a), (II)-(b), (III)-(d), (IV)-(c)

(2) (I)-(b), (II)-(a), (III)-(d), (IV)-(c)

(3) (I)-(a), (II)-(b), (III)-(c), (IV)-(d)

(4) (I)-(d), (II)-(c), (III)-(b), (IV)-(a)

74) Which of the following factors are favourable for the dissociation of oxyhaemoglobin in tissue ?

(1) High PO_2 and High PCO_2

(2) High PO_2 and Low H^+ concentration

(3) Low PCO_2 and High H^+ concentration

(4) High PCO_2 and Low PO_2

75) Match the column-I with column-II and choose correct option.

Column-I		Column-II	
(A)	TV	(i)	2300 mL
(B)	ERV	(ii)	1100 mL
(C)	IRV	(iii)	3000 mL
(D)	TLC	(iv)	500 mL

		(v)	4600 mL
		(vi)	5800 mL

(1) (A)-(iv), (B)-(ii), (C)-(iii), (D)-(v)

(2) (A)-(iv), (B)-(ii), (C)-(i), (D)-(vi)

(3) (A)-(iv), (B)-(ii), (C)-(iii), (D)-(vi)

(4) (A)-(iv), (B)-(i), (C)-(v), (D)-(vi)

76) Read the following statements -
(a) Pneumotaxic centre present in pons.
(b) Fibrinogen is required for blood clotting.
(c) O blood group is always Rh positive.
(d) Semilunar valve is present in all arteries.
(e) Pace maker is present in right atrium of human heart.
Select the option which includes only correct statements-

(1) (a), (b), (c)

(2) (b), (c), (d)

(3) (c), (d), (e)

(4) (a), (b), (e)

77) **Statement-I :-** In majority of nephrons, the loop of Henle is too short and extend only very little into the medulla.
Statement-II :- Glomerulus is a tuft of capillaries formed by afferent arteriole.

(1) Both statement-I & statement-II are true.

(2) Statement-I is true but statement-II is false.

(3) Both statement-I & statement-II are false.

(4) Statement-I is false but statement-II is true.

78) **Statement-I :** ADH helps in diluting the urine by acting on DCT and collecting duct.
Statement-II : Blood present in afferent and efferent arteriole will be different in amount of amino acid, glucose and urea.

(1) Both statement-I and II are true

(2) Both statement-I and II are false

(3) Statement-I is correct and statement-II is false

(4) Statement-I is false and statement-II is correct

79) Read the following statements carefully and choose the correct option :
(A) Glomerular filtration is a non-selective process that filters blood through three layers including podocytes, forming an ultrafiltrate.
(B) The ascending limb of Henle's loop impermeable to water and helps maintain the osmolar gradient by actively transporting electrolytes.
(C) ADH promotes diuresis by reducing water reabsorption from the collecting duct.
(D) ANF opposes the renin-angiotensin mechanism by causing vasodilation and reducing blood pressure.

Which of the following is correct ?

(1) Only A and B are true
(2) Only B, C and D are true
(3) Only A, B and D are true
(4) All four statements are true

80) Bilaterally symmetrical, triploblastic and acoelomate animals are :-

(1) Ctenophore
(2) Platyhelminthes
(3) Aschelminthes
(4) Mollusca

81) **Assertion :-** In coelomate animal body cavity is lined by mesoderm.
Reason :- In pseudocoelomate animal mesoderm is present as scattered pouches in between the ectoderm and endoderm.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.
(2) Both assertion and reason are correct and reason is not the correct explanation of assertion.
(3) Assertion is correct but reason is incorrect.
(4) Both assertion and reason are incorrect.

82) Cells forming the wall of blood capillaries are called

(1) Oxyntic cells
(2) Endothelial cells
(3) Parietal cells
(4) Haemocytes

83) Choose the incorrect one :-

(1) Albumin is not needed for BCOP.
(2) 6-8% part contributed by protein in plasma.
(3) Water contribute maximum part in plasma.
(4) RBC membrane are permeable for $HCO_3^{\ominus}$ and $Cl^{\ominus}$ ion

84) **Statement-I :-** Pumping pressure is the higher limit of blood pressure that shows the state of ventricle contraction.
Statement-II :- In man B.P. is measured in the brachial artery of arm.

(1) Both statement-I and statement-II are correct
(2) Both statement-I and Statement-II are incorrect
(3) Statement-I is incorrect and statement-II is correct
(4) Statement-I is correct and statement-II is incorrect

85) Tidal volume per minute is :

(1) 2500-3000 mL
(2) 1000-1100 mL
(3) 6000-8000 mL
(4) 1100-1200 mL

86) Choose the correct options-

	Column-I	**Column-II**
(1)	Inspiration	Diaphragm contraction
(2)	Expiration	EICM contraction
(3)	O_2 in Alveolar air	400 mm Hg
(4)	CO_2 in alveolar air	104 mm Hg

(1) 1
(2) 2
(3) 3
(4) 4

87) Which of the following defines net filteration pressure?

(1) BCOP – (GHP + CHP)
(2) (BCOP + GHP) – CHP
(3) GHP – (BCOP + CHP)
(4) (GHP – CHP) + BCOP

88)

The four structures listed are part of the human excretory system :
(1) Bladder (2) Kidney
(3) Ureter (4) Urethra
In which order does a molecule of urea pass through these structures :

	First ⟶			Last
(1)	1	2	3	4
(2)	1	4	3	2
(3)	2	1	3	4
(4)	2	3	1	4

(1) 1
(2) 2
(3) 3
(4) 4

89) Consider following four statements (A–D) and select the option which include all correct ones only :-
(A) Sweat produced by sweat gland contain urea, uric acid, creatine and sterols.
(B) Accumulation of urea in urine is called uremia.
(C) Analysis of urine help in clinical diagnosis of many metabolic disorders.
(D) On an average 25-30 milligrams of glucose is excreted out per day by a normal person.

(1) 'A', 'B' and 'C' only

(2) 'B' and 'C' only

(3) 'B' and 'D' only

(4) 'C' only

90) Which of the following structures are sensitive for CO_2 and H^+ concentration are -
A. Chemosensitive area
B. Chemoreceptor associated with aortic arch
C. Chemoreceptor associated with carotid artery
D. Carotid body
E. Aortic body

(1) Only A, B, E

(2) Only B, C, D

(3) Only A, C, E

(4) All A, B, C, D and E

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	2	1	1	4	2	3	1	2	2	3	3	4	1	2	4	3	1	2	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	2	2	2	4	2	2	3	3	1	2	2	2	1	3	2	2	2	3	1
Q.	41	42	43	44	45															
A.	3	4	4	1	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	3	4	1	1	4	1	3	2	3	4	1	2	4	4	2	1	4	1	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	3	2	4	4	3	2	2	2	4	4	3	4	4	1	3	3	1	4	4
Q.	86	87	88	89	90															
A.	2	4	2	3	1															

BIOLOGY

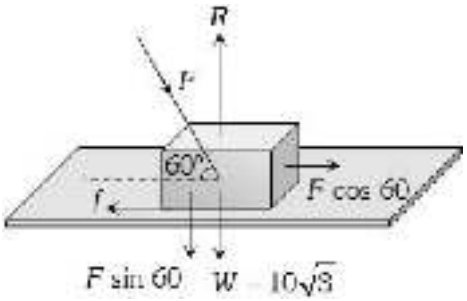
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	2	3	2	3	2	2	4	4	2	2	4	3	4	4	3	2	3	4	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	3	3	4	1	4	3	3	3	4	2	4	2	3	2	2	1	1	1	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	1	1	2	4	4	3	1	3	3	4	4	4	4	3	1	4	2	2	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	1	4	3	4	1	3	4	1	4	4	2	2	4	3	4	1	4	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	2	1	1	3	1	3	4	4	4										

SOLUTIONS

PHYSICS

1)

f = μR
F cos 60° = μ(W + F sin 60°)
Substituting $\mu = \frac{1}{2\sqrt{3}} \& W = 10\sqrt{3}$
We get F = 20 N



2) m = 1.76 kg
M = 25 m
= 25 × 1.76
= 44.0 kg
Mass of one unit has three significant figures and it is just multiplied by a pure number (magnified). So result should also have three significant figures.
The mass of 25 such balls is 44.0 kg.

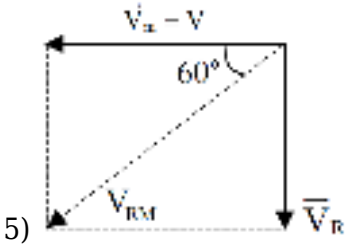
3) [A] = [L]
[B] = [k] = [T^{-1}]
[C] = [M°L°T°]

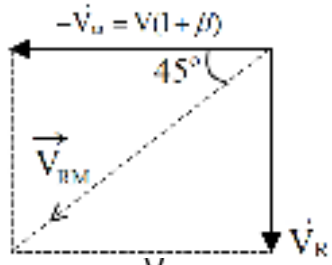
4)

Least count, L.C. = 1M.S.D. - 1V.S.D.
Lower least count means higher accuracy.
(A) 1 M.S.D. = 2mm
⇒ 10 V.S.D. = 9 M.SD
⇒ 1 V.SD = $\frac{9}{10}$ M.SD = 0.9 M.SD
L.C. = 1 M.S.D. – 0.9 MSD
= 0.1 M.S.D.
L.C. = 0.1 × 2 = 0.2 mm ____ (1)
(B) 1 M.S.D = 1 mm
⇒ 10 V.S.D. = 9 M.S.D.
⇒ 1 V.S.D. = 0.9 M.S.D.
 L.C. = 1 M.S.D. – 0.9 MSD
L.C. = 0.1 MSD

L.C. = 0.1 × 1 = 0.1 mm ___(2)
(C) 1 MSD = 2mm
20 VSD = 19 MSD
$1\ \text{VSD} = \frac{19}{20}\ \text{MSD} = \frac{19}{20} \times 2 = 1.9\ \text{mm}$ (1 MSD = 2mm)
 L.C. = 1 MSD – 1 VSD
L.C. = 1 × 2 mm – 1.9 mm
L.C. = 0.1 mm
(D) 1 MSD = 0.5 mm
10 VSD = 9 MSD
$1\ \text{V.S.D} = \frac{9}{10}\ \text{M.S.D.} = 0.9\ \text{MSD}$
= 0.9 × 0.5 = 0.45 mm
L.C. = 1MSD – 1 V.SD
L.C. = 1 × 0.5 mm – 0.45 mm
L.C. = 0.5 mm – 0.45 mm = 0.05 mm
Arranging in increasing order of accuracy (Smaller least count high accuracy)
⇒ A < B = C < D

5)





$\tan 60^\circ = \frac{V_R}{V_m}$

$\sqrt{3} = \frac{V_R}{V_m}$

$\sqrt{3} = \frac{V_R}{V}$ .....(1)

$\tan 45^\circ = \frac{V_R}{V(1+\beta)}$

$V(1+\beta) = V_R$

$V(1+\beta) = \sqrt{3}\,V$ [From (1)]

$\beta = 0.73$

6) $t = v^2 + 2v + 1$

$\frac{dt}{dt} = 2v\frac{dv}{dt} + \frac{2dv}{dt} + 0$

$1 = (2v+2)\frac{dv}{dt}$

$\frac{dv}{dt} = \frac{1}{2v+2}$

$\Rightarrow$ at t = 4 sec, v = 1 m/s and v = –3

$\Rightarrow \frac{dv}{dt} = \frac{1}{2+2} = \frac{1}{4} m/s^2$ and $\frac{dv}{dt} = -\frac{1}{4}$

7) $\frac{da^2}{dv} = \tan 37^\circ$

$2a\frac{da}{dv} = \frac{3}{4}$

$a\frac{da}{dt} \times \frac{dt}{dv} = \frac{3}{8}$

$a \cdot \frac{3}{x} \cdot \frac{1}{a} = \frac{3}{8}$

x = 8

8) Time taken by particles to collide

$t = \sqrt{\frac{2H}{g}}$

then $u\sqrt{\frac{2H}{g}} + V\sqrt{\frac{2H}{g}} = d$

$\Rightarrow \quad u + V = d\sqrt{\frac{g}{2H}}$

$\therefore V = d\sqrt{\frac{g}{2H}} - u$

9) x = 2m, u = –3m/s, a = 2m/s^2

(I) v = 0

$\Rightarrow$ u + at = 0

–3 + 2t = 0

$\Rightarrow t = \frac{3}{2} = 1.5s$

(II) v = 0, t = 3/2

(III) s = 0

$ut + \frac{1}{2}at^2 = 0$

$-3t + \frac{1}{2} \times 2t^2 = 0$

$t(-3 + t) = 0$

$\therefore t = 3 \sec$

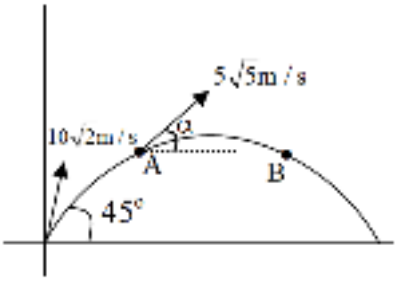
(IV) at t = 1.5sec., v = 0

$v^2 - u^2 = 2as$

$0 - 9 = 4s$

$\therefore s = -\frac{9}{4} m$

Hence it crosses the origin during the motion. $\Rightarrow x_{min} = 0$



10)

$10\sqrt{2}\cos 45^{o} = 5\sqrt{5}\cos\alpha$

$10 = 5\sqrt{5}\cos\alpha$

$\cos\alpha = \frac{2}{\sqrt{5}}$

$\sin\alpha = \frac{1}{\sqrt{5}}$

time taken in moving from A to B

$t = \frac{2\left(5\sqrt{5}\right)\sin\alpha}{g} = 1\,\text{sec}$

11) $V = \frac{dx}{dt}$

$\Rightarrow dt = \frac{dx}{v} = \frac{xdx}{9}$

$\int_0^t dt = \frac{1}{9}\int_5^{13} xdx$

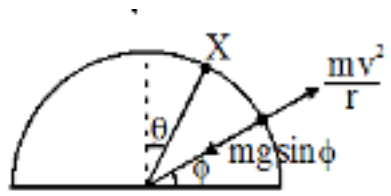
$t = \frac{1}{9} \times \frac{13^2 - 5^2}{2}$

t = 8 sec

12) $a_T = \frac{dV}{dt} = 4t + 1$

$a_C = \frac{V^2}{r}$

$a = \sqrt{a_c^2 + a_T^2} = 10\ \text{m/s}^2$



13)

$\frac{mv^2}{r} = mg\sin\phi$ .....(i)

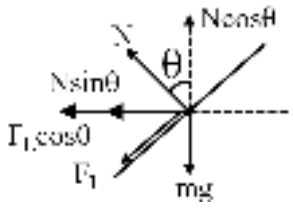
$mg\,r\cos\theta = \frac{1}{2}mv^2 + mg\,r\sin\phi$

$\frac{v^2}{rg} = 2\cos\theta - 2\sin\phi$ .....(ii)

$\sin\phi = 2\cos\theta - 2\sin\phi$

$3 \sin \phi = 2 \cos \theta$

$\sin \phi = \frac{2}{3} \cos \theta$

14) 

15)

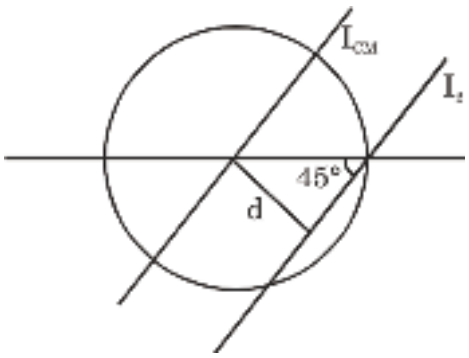
Let the angular velocity be ω, when the insect strikes the rod at C.
By the conservation of angular momentum about O.

$$\frac{MvL}{6} = \left[\frac{ML^2}{12} + M\left(\frac{L}{6}\right)^2\right]\omega$$

$$\frac{MvL}{6} = \left[\frac{ML^2}{12} + \frac{ML^2}{36}\right]\omega \quad \frac{MvL}{6} = \left[\frac{4ML^2}{36}\right]\omega$$

$$\frac{MvL}{6} = \left(\frac{ML^2}{9}\right)\omega$$

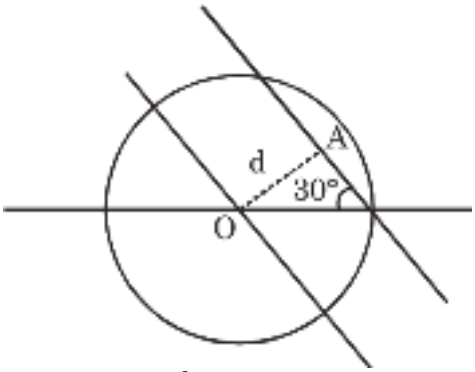
$$\Rightarrow \omega = \frac{3V}{2L}$$



16)

$I_2 = I_{CM} + Md^2$

$$= \frac{MR^2}{4} + M\left(\frac{R}{\sqrt{2}}\right)^2 = \frac{3}{4}MR^2$$



$I_1 = I_{CM} + Md^2$

$= \frac{MR^2}{4} + M\left(\frac{R}{2}\right)^2 = \frac{MR^2}{2}$

$\frac{I_2}{I_1} = \frac{3}{4}MR^2 \times \frac{2}{MR^2}$

$\Rightarrow \frac{I_2}{I_1} = \frac{3}{2}$

17)

$mg = B + F_V$

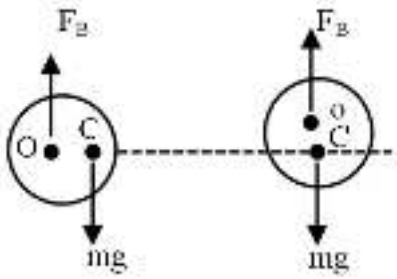
$mg = \frac{mg}{2} + F_V$

W.d. by mg = mgx

W.d. by $F_V = -\left(\frac{mg}{2}\right)x$

ratio = $\frac{-1}{2}$

18)



19) height balance additional pressure hence

$\rho g_s x = \rho\, g_{depth}\, y$

$g_s\, x = g_s(1 - d/R)y$

$\frac{x}{y} = 1 - \frac{d}{R}$

20) $w = \frac{1}{2}f_2\Delta\ell_2 - \frac{1}{2}f_1\Delta\ell_1$

$w = \frac{1}{2}\left[4 \times 10 \times 1 \times 10^{-3}\right]$

$-\frac{1}{2}[2 \times 10 \times 0.6 \times 10^{-3}]$
$= 14 \times 10^{-3}$ J

21)

On liquid force due to surface tension = $(2\pi R)\,T\cos\theta_C$
In equilibrium : force due to S.T = weight of rise liquid
$(2\pi R)T\cos\theta_C = mg$

$$T = \frac{\text{Weight}}{2\pi R} = \frac{6.2 \times 10^{-4}}{2 \times 3.14 \times 2 \times 10^{-3}} = 5 \times 10^{-2}\ \text{N/m}$$

22) $v_2A_2 = v_1A_1 \Rightarrow v_2 = v_1\frac{A_1}{A_2}$

$$\Rightarrow v_2 = v_1\frac{\pi r_1^2}{\pi r_2^2} = v_1\left(\frac{r_1}{r_2}\right)^2$$

$$= 0.25 \times \left(\frac{4}{1}\right)^2 = 4\ \text{m s}^{-1}$$

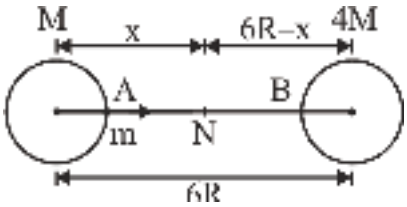
 horizontal range $x = v_2 \times t$

$$= v_2 \times \sqrt{\frac{2h}{g}} = 4 \times \sqrt{\frac{2 \times 1.25}{10}} = 4 \times \frac{1}{2} = 2\ \text{m}$$

23) $\Delta\vec{V} = (-2\hat{i} + 6\hat{j}) - (3\hat{i} + 4\hat{j})$

$= -5\hat{i} + 2\hat{j}$ unit vector $= \frac{-5\hat{i} + 2\hat{j}}{\sqrt{29}}$

24)



For neutral point

$$F_1 = F_2$$

$$\frac{GMm}{x^2} = \frac{G\ 4Mm}{(6R-x)^2}$$

$$\frac{1}{x} = \frac{2}{6R-x}$$

$$x = 2R$$

A/c to COME :-

$$E_A = E_N$$

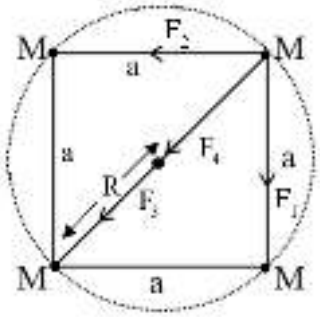
$$\frac{1}{2}mV^2 + \left(\frac{-GMm}{R}\right) + \left(\frac{-G4Mm}{5R}\right)$$

$$= 0 + \left(\frac{-GMm}{2R}\right) + \left(\frac{-G\,4Mm}{4R}\right)$$

$$V = \sqrt{\frac{3GM}{5R}}$$

25) NEW NCERT Pg # 136

As they are providing necessary centripetal force to each other $\Rightarrow \boxed{T_1 = T_2}$ now as $\omega = \frac{2\pi}{T}$ $\Rightarrow \boxed{\omega_1 = \omega_2}$ now $v = r\omega \Rightarrow v_1$ and v_2 are different due to different values of radii.



26)

$$\vec{F}_4 = \vec{F}_1 + \vec{F}_2$$

$$F_4 = \frac{\sqrt{2}GM^2}{a^2}$$

$$F_3 = \frac{GM^2}{2a^2}$$

$$F = \frac{GM^2}{a^2}\left(\frac{1}{2} + \sqrt{2}\right)$$

$$R = \frac{a}{\sqrt{2}}$$

$$a = \sqrt{2}R$$

$$\frac{GM^2}{\left(\sqrt{2}R\right)^2}\left(\frac{1}{2} + \sqrt{2}\right) = \frac{Mv^2}{R}$$

27) Shear force $F = T = mg = 0.020 \times 10 = 0.2$ N

Shear stress on the fluid $= \frac{F}{A} = \frac{0.2}{0.1} = 2$

Strain rate $= \frac{v}{\ell} = \frac{0.090}{0.30 \times 10^{-3}}$

$$\eta = \frac{\text{stress}}{\text{strain rate}} = \frac{2(0.30 \times 10^{-3})}{(0.090)}$$

$$= \frac{20}{3} \times 10^{-3} \text{ Pa s.}$$

28) The only force acting on the body is the viscous force

Here, $m\frac{vdv}{dx} = -6\pi\eta rv \qquad = -rv$

$$\Rightarrow \int_{v}^{0} m\,dv = \int_{0}^{x} -r\,dx \qquad \Rightarrow x = \frac{mv}{r}.$$

29)

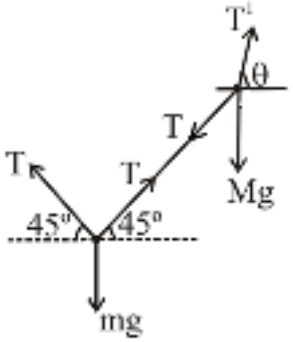
$$\Delta\ell_A = \Delta\ell_B$$

$$\Rightarrow \frac{F_A L_A}{\pi r_A^2 Y_A} = \frac{F_B L_B}{\pi r_B^2 Y_B}$$

$$\Rightarrow \frac{10 \times L}{\pi r_A^2 \times Y} = \frac{40 \times L}{\pi r_B^2 \times Y}$$

$$\Rightarrow \frac{r_A}{r_B} = \frac{1}{2}$$

Therefore other option are INCORRECT.



30)

$$\frac{T}{\sqrt{2}} + \frac{T}{\sqrt{2}} = mg$$

$$T = \frac{mg}{\sqrt{2}}$$

$$T' \cos\theta = \left(\frac{mg}{\sqrt{2}}\right)\frac{1}{\sqrt{2}}$$

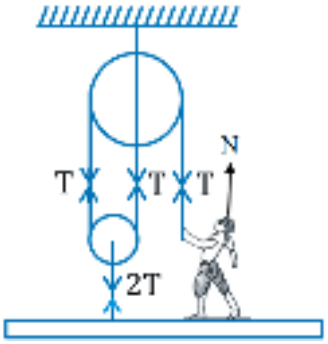
$$= T \cos 45^\circ$$

$$T' \sin\theta = Mg + \frac{mg}{\sqrt{2}}\frac{1}{\sqrt{2}}$$

$$= T \sin 45^\circ + Mg$$

$$\tan\theta = \frac{M + \frac{m}{2}}{\frac{m}{2}}$$

$$\tan\theta = 1 + \frac{2M}{m} = 1$$



31)

3T – 100g = 100 a

$$T = \frac{200 + 1000}{3} = \frac{1200}{3} = 400N$$

T + N – mg = ma (on men)

N = 60 × 2 + 600 – T

N = 720 – 400 = 320 N

32)

[NCERT pg # 113]

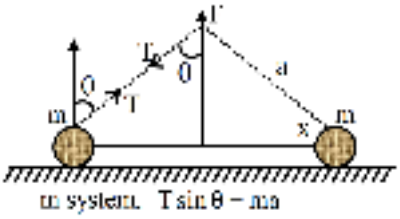
max acceleration of block = μg

= 2 m/s^2

$S_{relative} = (a_{relative}\ t^2/2)$

$$t = \sqrt{\frac{2 \times 5}{(3-2)}} = \sqrt{10} \text{ sec}$$

$$s_{trolley} = 0 + \frac{1}{2} \times 3 \times \left(\sqrt{10}\right)^2 = 15 \text{ m}$$



33) m system $T \sin\theta = ma$

m system, T sin θ = ma ...(1)

$$2T \cos\theta = F \Rightarrow T = \frac{F}{2\cos\theta} \quad ...(2)$$

using (2) in (1), F tan θ = 2ma

$$a = \frac{F\tan\theta}{2m} = \frac{F}{2m}\frac{x}{\sqrt{a^2 - x^2}}$$

34) $\frac{du}{dx} = \frac{4x^3}{4} - \frac{2x}{2} = 0$

$x(x^2 - 1) = 0$

$x = 0, 1, -1$

$$\frac{d^2u}{dx^2} = 3x^2 - 1$$

x = 0, $\frac{d^2u}{dx^2}$ = –ve

$x = \pm 1, \quad \frac{d^2u}{dx^2} =$ +ve

$\Rightarrow$ U_{min} when $x = 1$

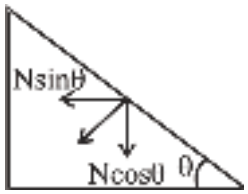
$\Rightarrow$ $U_{min} = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$J

$TE = U_{min} + K_{max}$

$2 = -\frac{1}{4} + K_{max} \Rightarrow K_{max} = 9/4$ J

$\frac{1}{2} \cdot 1 \cdot v^2 = \frac{9}{4}$

$v^2 = \frac{9}{2}$ $\qquad$ $v_{max} = \frac{3}{\sqrt{2}}$ m/s

35) 

$N\sin\theta = kx$

$\frac{kx}{mg} = \tan\theta \Rightarrow x = \frac{mg\tan\theta}{K}$

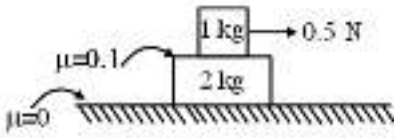
$N\cos\theta = mg$

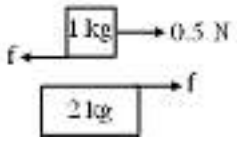
$v = \frac{1}{2}kx^2 = \frac{km^2g^2\tan^2\theta}{k^2} = \frac{m^2g^2\tan^2\theta}{2k}$

36) $P = m\frac{du}{dt}v$

$v = \sqrt{\frac{2pt}{m}}$

$= \sqrt{\frac{2 \times 0.5 \times 5}{0.2}} = 5$ m/s



37) 

$\frac{5}{30}$

$0.5 = 3a \Rightarrow a = \frac{5}{30}$

$f = 2a = 2\left(\frac{5}{30}\right) = \frac{1}{3}N$

$f_{max} = 0.1 \times 10 = 1N$

$\Rightarrow$ $f = \frac{1}{3}N$ will act

Work done by lower block

$= -f(s)$

= –1/3 (3) = –1 J

38) $\frac{\Delta Q}{Q} = \frac{2\Delta X}{X} + \frac{1}{2}\frac{\Delta Y}{Y}$ $\Delta Q = Q\left(\frac{2\Delta X}{X} + \frac{1}{2}\frac{\Delta Y}{Y}\right)$

$= \frac{2X}{\sqrt{Y}}\Delta X + \frac{X^2\Delta Y}{2Y\sqrt{Y}}$

$= \frac{(4XY)\Delta X + X^2\Delta Y}{2Y^{3/2}}$

39)

Percentage error $= \frac{\text{Least count}}{\text{Measured value}} \times 100$

Percentage error = $\frac{0.01}{5} \times 100 = 0.2\%$

Thus , the percentage error is 0.2%.

40) Limiting friction

$f_L = \mu_S$ mg

$200 = \mu_S \times 500$

$\mu_S = \frac{2}{5} = 0.4$

Kinetic friction

$f_K = \mu_K$ mg

$150 = \mu_K \times 500$

$\mu_K = \frac{3}{10} = 0.3$

41) The momentum of the two-particle system, at t = 0 is

$\vec{P}_1 = m_1\vec{v}_1 + m_2\vec{v}_2$

Collision between the two does not affect the total momentum of the system.

A constant external force $(m_1 + m_2)g$ acts on the system.

The impulse given by this force, in time t = 0 to t = $2t_0$ is $(m_1 + m_2)g \times 2t_0$

 Change in momentum in this interval

$= \left|m_1\vec{v}_1' + m_2\vec{v}_2' - (m_1\vec{v}_1 + m_2\vec{v}_2)\right|$

$= 2(m_1 + m_2)gt_0$

42) $v = \sqrt{\frac{GM}{R+h}}$

$v' = 2\sqrt{\frac{GM}{R+h}}$

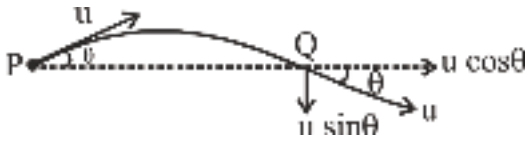
 it will escape.

43) $\frac{-GMm}{R} + KE = \frac{-GMm}{2R} + \frac{1}{2}m\frac{GM}{2R}$

$$KE = \frac{3}{4}\frac{GMm}{2R} = \frac{3}{4}mgR$$

44)

$$\vec{\tau}_{AV} = \frac{\vec{L}_f - \vec{L}_i}{\text{time}}$$



$\vec{L} = 0$, $\vec{L}_f = m(u \sin \theta)R$

$$\text{Time} = \frac{R}{u\cos\theta}$$

So $\tau_{AV} = \dfrac{m(u\sin\theta)R}{R/u\cos\theta}$ $\tau_{AV} = \dfrac{mu^2 \sin 2\theta}{2}$

45) Acceleration $\dfrac{\sum F}{M} = \dfrac{F}{M}$
(friction force is zero)

CHEMISTRY

46) Moles = 5/100 = 0.05.
Volume at STP = 0.05 × 22.4 = 1.12 L

47)

Number of atoms = mole × N_A × Atomicity

48) Conc. decreases as volume is increased.

49) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$. ΔH
= [– 393.5 + 2(– 285.8)] – (–74.8)
= – 890 kJ/mol

50) Limiting reagent = 0.20 mol NaOH. Heat = 0.20 × 57 = 11.4 kJ

51) Kp depends only on temperature, not P or x.

52) Min. M. wt = (78.4 × 100)/0.5 = 15, 680 = 1.568×10^4

53) Using n + ℓ value n + ℓ(3d) > n + ℓ(4s).

54)

Zero

55) Reverse : K = $\frac{1}{40}$, Divide by 2 : K = $\sqrt{\left(\frac{1}{40}\right)} = \frac{1}{\sqrt{40}}$

56) $\bar{\nu} = R_H \times 1^2 \left[\frac{1}{2^2} - \frac{1}{n_2^2}\right]$

 for $n_2 = 3$; $\bar{\nu} = \frac{5R}{36}$

$n_2 = 4$, $\bar{\nu} = \frac{3R}{16}$

$n_2 = 5$, $\bar{\nu} = \frac{21R}{100}$

57) In N^{3-} nitrogen is in its lowest o.s.

58) $S = \frac{w \times 10}{M}$

$Ca_3(PO_4)_2(s) \rightleftharpoons 3Ca^{2+}(aq) + 2PO_4^{3-}(aq)$

3s 2s

$S = \frac{w \times 100}{M \times 100} = \frac{w \times 10}{M}$

$Ksp = (3s)^3 (2s)^2$

$= 108 \times 10^5 \times \left(\frac{w}{M}\right)^5$

$= 1.08 \times 10^7 \times \left(\frac{w}{M}\right)^5$

59) $HA \rightleftharpoons H^+ + A^-$

C–Cx Cx Cx

$pH = pKa + \log \frac{[A^-]}{[HA]}$

$pH = pKa + \log\left(\frac{Cx}{C - Cx}\right)$

$pH = pKa + \log\left(\frac{x}{1-x}\right)$

$pH - pKa = \log\left(\frac{x}{1-x}\right)$

60) $d^8 \rightarrow Ni^{+2}$, $d^7 \rightarrow Ni^{+3}$

Total charge of nickel

$(0.96 \times 2) + (0.04 \times 3) = 2.04$

Number of O^{2-} ion = $\frac{2.04}{2} = 1.02$

 $NiO_{1.02}$
$Ni_{0.98}$ $O_{1.00}$

61) $\Delta KE = x - \frac{x}{9} = \frac{8x}{9}$
$\Delta PE = 2 \Delta KE$
$= \frac{2 \times 8x}{9}$
$= \frac{16x}{9}$

62) $\frac{\frac{1}{\lambda_{min}} = R\left(\frac{1}{4} - \frac{1}{\infty}\right)}{\frac{1}{\lambda_{max}} = R\left(\frac{1}{4} - \frac{1}{9}\right)}$
$\frac{\lambda_{min}}{\lambda_{max}} = \frac{5}{9}$

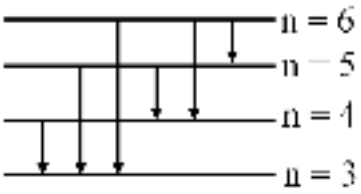
63) $C_4H_{10} + \frac{7}{2}O_2 \rightarrow 4CO_2 + 5H_2O$
(g) (g) (g) ( )
$\Delta n_g = 4 - 7.5$
$= -3.5$
$\Delta H = \Delta E + \Delta n_g RT$
$|\Delta H| < |\Delta E|$

64) % of H = $\frac{7.5}{116} \times 100 = 6.5$
% of O = $\frac{60}{116} \times 100 = 51.7$
% of C = $\frac{48.5}{116} \times 100 = 41.8$
 H : O : C
$\frac{6.5}{1} : \frac{51.7}{16} : \frac{41.8}{12}$
6.5 : 3.25 : 3.5
Emperically formula is approx CH_2O

65) $\lambda = \frac{h}{m\nu}$

66) For isothermal process
$\Delta U = 0$

67) $\overset{+7}{KMnO_4} \rightarrow \overset{+2}{MnCl_2}$
O.A

68) 

69) Formation from elements in standard states
$C(graphite) + O_2(g) \rightarrow CO_2(g)$

70) $\Delta H = \Delta U + \Delta n_g RT$

$= -726 - \frac{1}{2} \times \frac{8.3 \times 300}{1000}$

$= -727.245$

71) $\Delta G < 0$

$\Rightarrow \Delta H - T\Delta S < 0$

$\Rightarrow T > \left(\frac{\Delta H}{\Delta S}\right) = T_e$

$\Rightarrow T > T_e$

72) $M_1V_1 = M_2V_2$

$0.1 \times V = M_2 \times 2V$

$M_2 = 0.05$

$pH = -\log 0.05$

$= 1.3$

73)

$X^{th} = (n-1)s^2p^6d^{10}$

$Y^{th} = (n-1)\,s^2\,p^6\,d^{10} + ns^2$

at . No. = same. $\boxed{n > m}$

74) $\sigma^2_{1s}, \sigma^{x2}_{1s}, \sigma^2_{2s}, \sigma^{x2}_{2s}, \sigma^2_{2p_z}, [\pi^2_{2px} = \pi^2_{2py}]$

$\left[\pi^{*1}_{2px} = \pi^{*1}_{2py}\right]$

Homo-Highest occupied molecular orbital.

75)

If s-p mixing absent then configuration $\sigma 1s, \sigma^*1s, \sigma 2s, \sigma^*2s, \sigma 2p_z, [\pi(2p_x) = \pi(2p_y)], [\pi^*(2p_x)' = \pi^*(2p_y)]$

76) $ICl_2^+ = 2LP + 2BP = 4$

$ICl_2^- = 3\ LP + 2BP = 5$

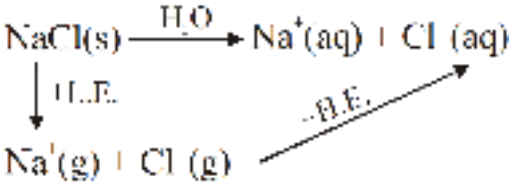
ICl_4^- = 2 LP + 4 BP = 6

77) Bond strength ∝ Bond order
removal of electron from antibonding MO increases B.O.
NO & O_2 has valence e^- in π* orbital.

78)

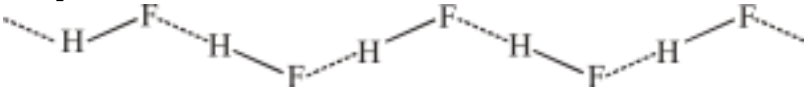
XeF_2 = sp^3d – Linear geometry.
N_3^- → N ≡ N → N^- – sp– Linear
PCl_5(s) – PCl_6^- – sp^3d^2 – square bi pyramidal anion.
$I_2Cl_6 \rightarrow ICl_2^+ + ICl_4^-$
↓
sp^3 (2LP)

79)



When ionic solid is dissolved in H_2O, attraction between cation and anion is broken (by giving L.E.) and attraction between ion and solvent molecules takes place and solvation energy is released.

80) Assertion is incorrect since in H-bonding, dipole-dipole interaction are non covalent. But Ion-dipole interaction can also result in H-bond formation.
Reason is correct, F is most electronegative element & in HF structure symmetrical H-bonds are present.

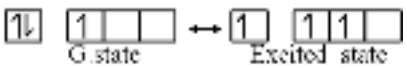


(H-bonding in HF)

81)

Explanation : Asking about VBT molecule got stability.
According to VBT. in covalent bond formation only those orbital having unpaired e^- participates.

82)

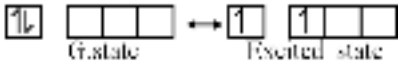
(i) B = $1s^2$, $2s^2$, $2p^1$ = 

(ii) N = $1s^2$, $2s^2$ = $2p^3$ = 

– No excitation is possible.

(iii) P = [Ne] $3s^2 3p^3$ = 

– 'P' shows 3 valency in ground state.

(iv) Be = 

83)

Addition of 2nd el^- is always endothermic

84)

Lower n value experience more attraction force then higher n value el^-.
Hense more energy is required.

85)

having lower IE value is a metal.

86)

Elements in the same *group* have the same number of valence electrons. Elements in the same *period* have the same valence shell, but the number of valence electrons increases.
Final Answer: Option (2)

87)

Across a period, increasing nuclear charge pulls electrons closer to the nucleus, reducing atomic size.

88)

Sc < y < La ≃Ac. because lanthanoid contraction will be observed after La not in La.

89)

IE of anion is decided by EA of neutral atom. EA of Cl > EA of F.

90) NCERT Pg. No. # 18

BIOLOGY

91)

NCERT (XIth) Page No. # 159

92) NCERT (XIth) Page No. # 112, 113, 114, 115, 116

Statement II & III are incorrect.

93) NCERT (XIth) Page No. # 111 & 112

94) NCERT-XI Pg. # 151

95)

NCERT (XIth) Page No. # 112, Fig. 9.3

96)

NCERT-XI, Pg. No. # 97, 98

97) NCERT-XI, Pg. # 100, 101 & 102

98)

NCERT Pg. # 89

99) NCERT XI Pg. # 98, 100

100) NCERT XI Pg. # 97, 98

101) NCERT-XI, Pg. # 99

102) NCERT (XIth) Page No. # 89, 90, 91, 92

103)

NCERT (XIth) Pg. # 101

104) NCERT Pg. No. 98, 99

105) Module Pg. # 162

106) NCERT (XIth) Pg. # 126, 127

107) NCERT (XIth) Pg. # 126

108) NCERT XI, Pg.No. # 122, 123, 124

109)

NCERT (XIth) Pg. # 122, 124, 128

110)

Solution
Each daughter cell has 8 chromosomes, meaning the haploid number (n) is 8. In prophase I, homologous chromosomes pair up to form bivalents. Therefore, the number of bivalents in the mother cell will be equal to the haploid number, which is 8.

Final answer: Option 1

111)

Solution
A bivalent has two homologous chromosomes.
Each chromosome has two sister chromatids, and each chromatid has its own kinetochore.
Therefore, total kinetochores = 15 bivalents × 2 chromosomes × 2 kinetochores per chromosome = 60.
Final answer : Option 2

112)

Module

113) NCERT XI
Module # 1

114) Old NCERT (XIth) Page # 4

115) NCERT-XI, Pg # 8

116)

NCERT (XIth) Page No. # 12,13, 90

117)

NCERT (XIth) Page No. # 14, Fig. 2.3
NCERT (XIth) Page No. # 15, Fig. 2.4
NCERT (XIth) Page No. # 17, Fig. 2.5
NCERT (XIth) Page No. # 20, Fig. 2.6

118)

NCERT (XIth) Page No. # 17, 18

119)

NCERT XI(E)/(H) Pg # 13 (2.1.2)

120) NCERT (XIth) Page No. # 15

121)

NCERT (XIth) Page No. # 16

122)

NCERT (XIth) Page No. # 21

123)

NCERT Pg # 17, 18

124)

NCERT-XI, Pg # 21

125) NCERT (XIth) Page No. # 16, 17

126)

NCERT (XIth) Page No. # 13

127)

NCERT (XIth) Page No. # 13

128)

NCERT (XIth) Page No. # 33, Module-1

129) NCERT XI Pg. # 32

130) NCERT XI page 29

131) NCERT XI Pg. # 33

132) NCERT Pg. # 29, 30

133) NCERT (XIth) Page No. # 32, 33

134) NCERT Pg. # 30, 32

135)

NCERT (XIth) Page No. # 24,25,26,27

136)

NCERT XI Pg. 47, 48

137) NCERT Page. No : 46

138) NCERT, Pg. # 41

139)

NCERT XI Pg. 38

140) NCERT, Pg. # 45

141) NCERT XI Pg. 40, 41

142) NCERT-XI, Pg. # 49

143) NCERT-XI, Pg. # 48, Para 4.2.11.4

144) NCERT-XI, Pg. # 48, Para 4.2.11.4

145)

NCERT XI Pg. 44, 45

146)

NCERT XI Pg. 103

147)

NCERT XI (old) Pg. 103

148) NCERT Pg. # 102, 103 (Old)

149)

NCERT XI (old) Pg. 103

150) NCERT-XII, Pg. # 113

151)

NCERT XI (old) Pg. 113

152)

NCERT XI (old) Pg. 112

153) NCERT Pg # 82

154)

NCERT XI (old) Pg. 82

155) NCERT XI Pg. No. # 48

156)

NCERT XI Pg. No. # 80, 81

157) NCERT Pg#201

158) NCERT Page No. 199

159) NCERT Page No. 195

160)

NCERT Based

161)

NCERT Based

162) NCERT, Pg. # 201

163) NCERT Pg.# 186,187.

164) NCERT Pg.#189.

165) NCERT Pg.#186, 187.

166) NCERT Pg.#190, 195, 198.

167) NCERT, Pg. # 207, 208

168) NCERT-XI, Pg, # 208, 209

169)

NCERT, Pg. # 208-212

170)

NCERT Pg. # 42

171) NCERT, Pg. # 39

172)

NCERT Pg. # 202

173) NCERT Pg. # 193,194

174) Module

175) NCERT-Pg : 271

176) NCERT Pg # 185, 187

177)

NCERT Pg. # 209

178)

NCERT Pg. # 212

179)

NCERT Pg. # 213

180) NCERT XI, Pg. # 190